ABCB

BCA Section J - Assessment and Verification of an Alternative Solution

2010

Handbook

NON-MANDATORY DOCUMENT

INFORMATION HANDBOOK

BCA Section J - Assessment and Verification of an Alternative Solution

2010

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The Australian Building Codes Board (ABCB) and the participating Governments are committed to enhancing the availability and dissemination of information relating to the built environment. Where appropriate, the ABCB seeks to develop non-regulatory solutions to building related issues.

This Handbook on BCA Section J Assessment and Verification of an Alternative Solution (the Handbook) is provided for general information only and should not be taken as providing specific advice on any issue. In particular, this Handbook is not mandatory or regulatory in nature. Rather, it is designed to assist in making information on this topic readily available.

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Preface

The Inter–Government Agreement (IGA) that governs the Australian Building Codes Board (ABCB) places a strong emphasis on reducing reliance on regulation, including consideration of non–regulatory alternatives such as non–mandatory documents.

This Handbook is one of a series produced by the ABCB. This series of Handbooks is being developed in response to comments and concerns expressed by government, industry and the community that relate to the built environment. ABCB Handbooks are informative non-mandatory documents containing generic advice on factors that may be considered, or approaches that may be taken, in dealing with specific building issues. Any numerical values or specific instructions contained in this Handbook should be considered as examples of outcomes from the proposed process rather than specific guidance on the issues. It should be noted that the Handbook represents the views of the ABCB and the authors and there may be other equally valid points of view on these topics.

The application and interpretation of the Building Code of Australia (BCA) Assessment and Verification Methods is the responsibility of the Building Control Authority's in the States and Territories. This Handbook has been produced to provide guidance in generic terms. The Administrations in the States and Territories may also produce advisory documents.

It is expected that this Handbook will assist a broad range of stakeholders to understand the application of the BCA energy efficiency provisions. It will also assist in developing appropriate building solutions by better understanding the requirements of the BCA and the powers that the Building Control Authority has under building legislation.

Energy efficiency provisions are relatively new to the BCA and industry has sought guidance on how the provisions are applied. Although each project is unique, and each jurisdiction has different requirements in its building legislation, this Handbook provides some advice on general principles.

To make comment on this Handbook or to seek further information please contact the General Manager of the ABCB at e-mail address abcb.office@abcb.gov.au or mail address GPO Box 9839 Canberra ACT 2601.

Acknowledgments

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- The States and Territory Administrations
- Dr Ernest Donnelly of Pacific and Australia Consulting Engineers
- Mr Murray Mason of ACADS-BSG
- Mr P.C. Thomas of Team Catalyst

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1. Introduction

Reminder:

This Handbook is not mandatory or regulatory in nature and compliance with it will not necessarily discharge a user's legal obligations. This Handbook should only be read and used subject to, and in conjunction with, the general disclaimer at page i.

This Handbook also needs to be read in conjunction with the building legislation of the relevant State or Territory. It is written in generic terms and it is not intended that the content of the Handbook counteract or conflict with the legislative requirements, any references in legal documents, any handbooks issued by the Administration or any directives by the Building Control Authority. See Appendix A for Summary of State and Territory advice.

1.1 Background of Energy Efficiency in the BCA

Energy efficiency provisions were introduced into the Building Code of Australia (BCA) in stages. The first was in 2003 for Class 1 and 10 buildings (BCA Volume Two Housing Provisions). This was followed in 2005 by provisions in Volume One for Class 2 buildings (apartments) and 3 buildings (hotels, motels dormitories etc) and Class 4 parts of buildings (residences over other buildings). The range of buildings became complete in 2006 when provisions for Classes 5 to 9 buildings (all other applications) were added to Volume One. At the same time, the provisions for Classes 1 and 10 in Volume Two were made more stringent. In 2010 the stringency of the provisions in both Volumes was again increased. Note that these dates were when the provisions were introduced into the national BCA and not necessarily when the States and Territories adopted them into building law. For this information refer to the History of Adoption at the back of the BCA.

1.2 Purpose of the Handbook

This Handbook is targeted at designers, energy analysts, mechanical engineers, electrical engineers and other specialist designers who are familiar with energy analysis software used to model the annual energy consumption of a building. The intent is to familiarise them with the BCA, the intent of a specific clause and how it should be interpreted.

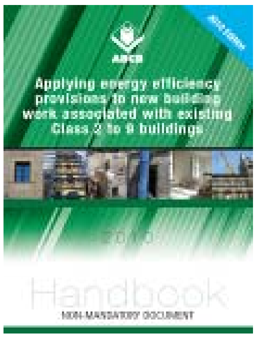
The requirements of the BCA relating to energy efficiency represent the minimum acceptable building standards as determined by wide consultation with governments, industry and the community.

This Handbook needs to be read in conjunction with the building legislation of the relevant State or Territory. It is written in generic terms and it is not intended that the content of the Handbook counteract or conflict with the legislative requirements, any references in legal documents, any handbooks issued by the Administration or any directives by the Building Control Authority.

This Handbook does not override or replace the BCA, but rather provides additional information and guidance with the principles explained. It also uses examples to aid the user in the application of the new BCA energy efficiency provisions associated with an existing building. It is recommended that users of this Handbook seek specialist advice in its application to specific projects.

It also needs to be kept in mind that the requirements of the BCA represent the minimum acceptable standard as determined by wide consultation with governments, industry and the community.

1.3 Other Reference Material



In addition to the BCA and the Guide to Volume One, the ABCB has also produced a Handbook titled "Applying Energy Efficiency Provisions to New Building Work Associated With Existing Class 2 to Class 9 Buildings", another titled "Energy Efficiency Provisions for BCA 2010 Volume One" and three relevant education modules, namely:

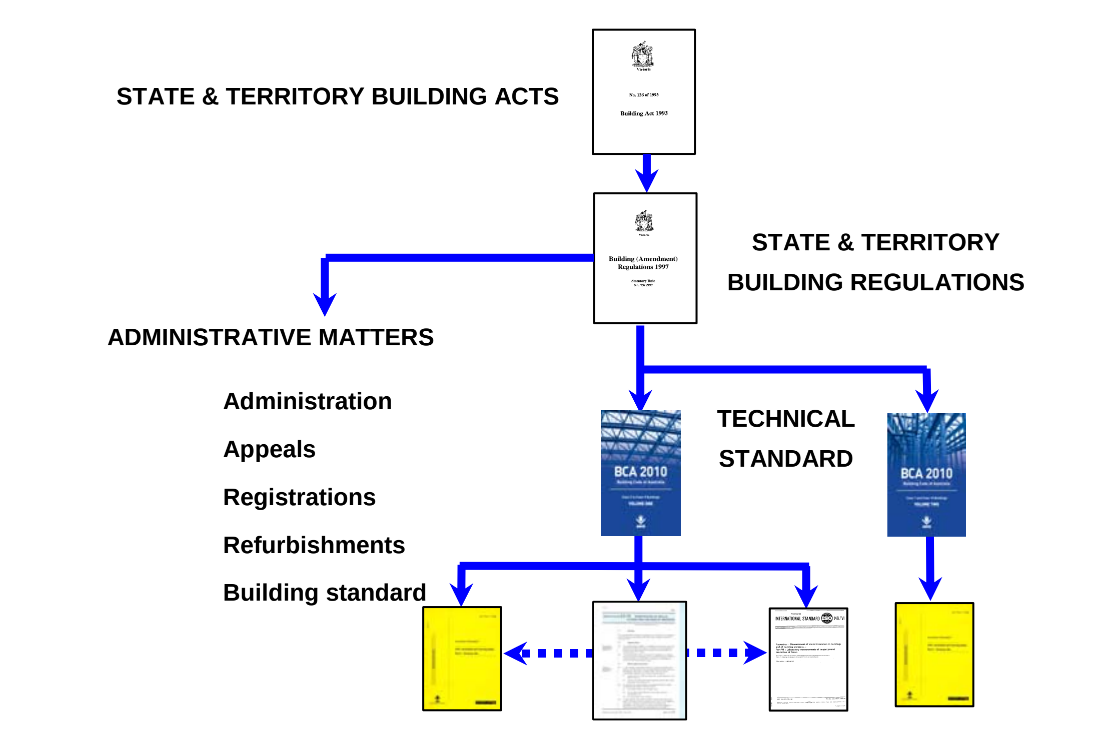
- BCA Resource Kits Module One – An Introduction to the Building Code of Australia.
- BCA Resource Kits Module Two – Understanding the BCA's Performance Requirements.
- BCA Resource Kits Module Four – Understanding Energy Efficiency Provisions for Class 2 to 9 Buildings.

1.4 Building Control

Building control is the responsibility of each State and Territory. The BCA is adopted via the State and Territory building legislation as the technical standard for the design and construction of buildings. The legislation generally applies the BCA to new buildings, new building work in existing buildings and changes in building classification or use.

Typically, the legal arrangement is such that the BCA contains the technical provisions, while the building administrative procedures are contained in an Act and usually supported by associated Regulations. Refer to **Figure 1** for an overview of the legislative hierarchy.

Figure 1 - Legislation hierarchy

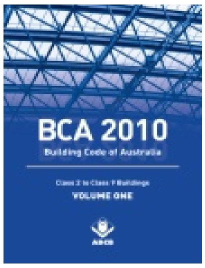


1.5 The BCA

The BCA is produced and maintained by the Australian Building Codes Board (ABCB) on behalf of the Australian Government and State and Territory Governments. The BCA has been given the status of building regulations by all States and Territories.

The goal of the BCA is to enable the achievement of nationally consistent, minimum necessary standards of relevant, health, safety (including structural safety and safety from fire), amenity and sustainability objectives efficiently.

This goal is applied so–



- there is a rigorously tested rationale for the regulation; and
- the regulation generates benefits to society greater than the costs (that is, net benefits); and
- the completive effects of the regulation have been considered and the regulation is no more restrictive than necessary in the public interest; and
- there is no regulatory or non-regulatory alternative that would generate higher net benefits.

The BCA contains technical provisions for the design and construction of buildings and other structures, covering such matters as structure, fire resistance, access and egress, services and equipment, and energy efficiency as well as certain aspects of health and amenity.

Defined terms

The BCA contains many definitions in Part A1 Interpretation. These definitions are shown in the text of the BCA in italics. When readers come across a defined terms they should refer to the definition as it may be very different from what many be considered common usage or what a dictionary contains. The definitions are specifically tailored for the BCA context. Fore Example, a "*conditioned space"* is very specifically defined as-

"**Conditioned space** means a space within a building, including a ceiling or under-floor supply air plenum, where the environment is likely, by the intended use of the space, to have its temperature controlled by *air-conditioning*, but does not include-

(a) a non-habitable room of a Class 2 building or Class 4 part of a building in which a heater with a capacity of not more than 1.2 kW or 4.3 MJ/hour provides the air-conditioning; or

(b) a space in a Class 6, 7, 8 or 9b building where the input energy to an *air-conditioning* system is not more than 15 W/m^2 or 15 $J/s.m^2$ (54 $KJ/hour.m^2$).

This is very specific and may be quite different from what practitioners may expect. In particular it exempts some specific applications even though they would otherwise be considered to be air-conditioned.

Building Classifications

Part A3 of the BCA contains descriptions of the various building classifications, such as a

Class 5 being "an office building used for professional or commercial purposes, excluding buildings of Class 6, 7, 8 and 9". Where there is doubt as to a buildings classification the Building Control Authority should be consulted.

1.6 The BCA Performance Requirements

The Objectives, Functional Statements and Performance Requirements form the BCA performance hierarchy. The Performance Requirements have been developed to satisfy both the BCA Objectives and Functional Statements.

The Objectives and Functional Statements are provided as guidance. The Performance Requirements are a mandatory component of the BCA. The DTS Provisions are typical solutions that comply.

A brief summary of each element is as follows-

- the Objectives describe the community expectations for buildings;
- the Functional Statements describe how buildings achieve the objectives; and
- the Performance Requirements outline the level of performance which must be met by building materials, components, design factors and construction methods in order for a building to meet the Objectives and Functional Statements. The Performance Requirements are generally qualitative.

Compliance with the Performance Requirements is achieved by using a Building Solution. Refer to **Figure 2**.

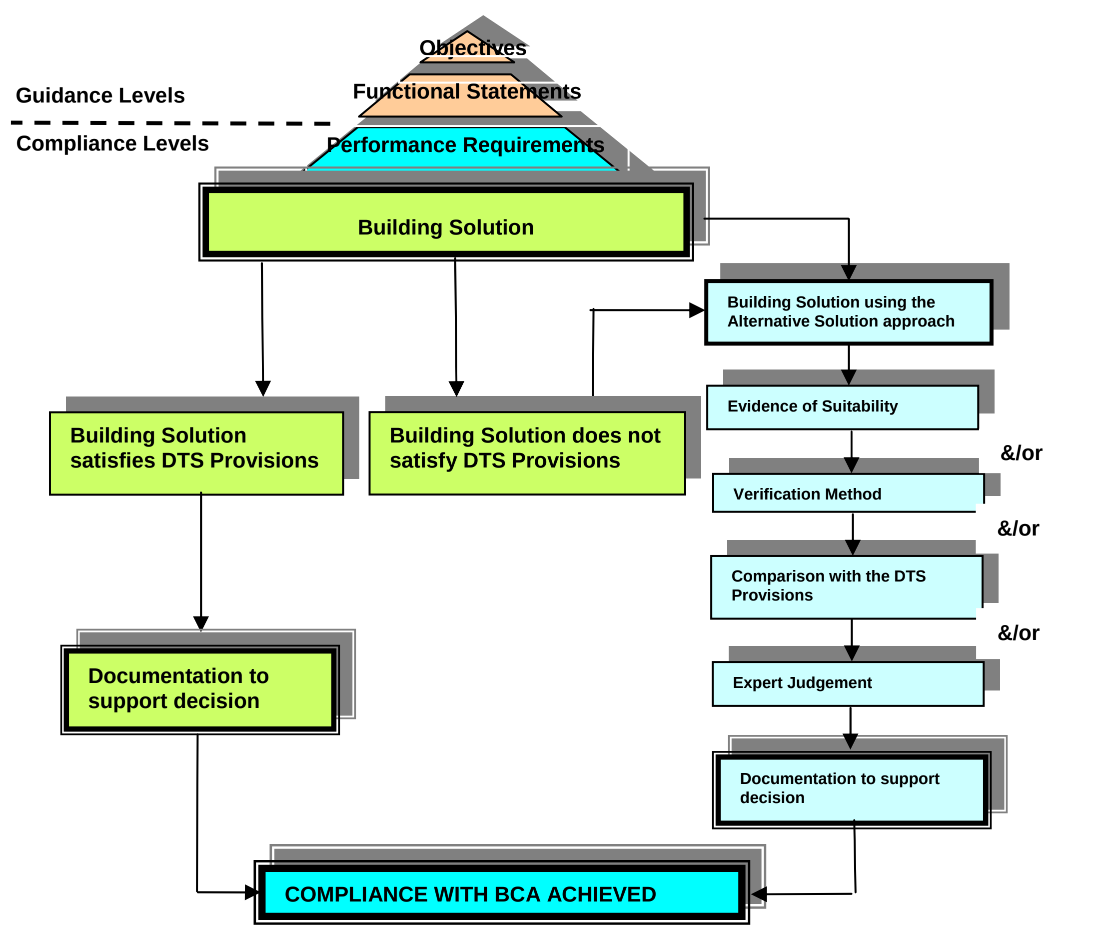
The Objective for energy efficiency is JO1 and reflects the governments' goal of reducing greenhouse gas emissions attributed to buildings.

There are essentially three options for a Building Solution-

- compliance with the DTS Provisions (for energy efficiency that is Parts J1 to J8);
- use of an Alternative Solution justified by an appropriate Assessment Method; or
- a mixture of both DTS Provisions and Alternative Solutions.

Essentially the DTS Provisions, prescribed in the BCA are only one type of Building Solution that will meet the Performance Requirements. The other type of Building Solution that will meet the Performance Requirements is an "Alternative Solution". An Alternative Solution is a Building Solution that is outside the DTS Provisions yet is found to be compliant with the Performance Requirements using one of four Assessment Methods as shown in **Figure 2.** This figure outlines the performance based compliance framework.

Figure 2 – Performance based compliance framework



1.7 How to comply with the Performance Requirements

The following are the relevant Clauses within the BCA that describe how the Performance Requirements are used.

Part A0 is the part in the BCA which outlines the structure of the BCA and specifies the compliance parameters of the BCA. **Clause A0.4** states that a Building Solution will comply with the BCA if it meets the Performance Requirement.

Clause A0.5 states that compliance to the Performance Requirements is achieved by either complying with the DTS Provisions or formulating an Alternative Solution that is shown to meet the Performance Requirement, or a combination of both (DTS solution and Alternative Solution).

Clause A0.8 states that an Alternative Solution must be assessed by an appropriate Assessment Method.

Clause A0.9 specifies the available Assessment Methods.

1.8 Assessment Methods and Verification Methods

When an Alternative Solution is used, it must be proven that it meets the applicable Performance Requirements. In order to do this, the Alternative Solution must be assessed by using at least one of the specified Assessment Methods outlined in BCA Clause A0.9.

The Assessment Methods are as follows. These are also depicted in **Figure 2**.

(a) Evidence to support that the use of a material, form of construction or design meets a Performance Requirement or DTS Provision as described in Clause A2.2.

(b) Verification Methods such as-

(i) the Verification Method in the BCA; or

(ii) other Verification Methods that the Building Control Authority accepts for determining compliance with the Performance Requirements.

(c) Comparison with the DTS Provisions.

(d) Expert Judgement.

1.8.1 Assessment Method (a) - Evidence of Suitability

Assessment Method One can be used for both the DTS Provisions and an Alternative Solution. It is located in Part A2 Clause A2.2.

Evidence of suitability can generally be used to support a material, form of construction or design that satisfies either a Performance Requirement or a DTS Provision.

The form of evidence that may be used consists of one, or a combination of, the following methods:

- A report from a Registered Testing Authority.
- A Certificate of Conformity or a Certificate of Accreditation.
- A certificate from a professional engineer or other appropriately qualified person.
- A current certificate issued by a product certification body that has been accredited by the Joint Accreditation System of Australia and New Zealand (JAS-ANZ).

- Any other form of documentary evidence that adequately demonstrates suitability.

1.8.2 Assessment Method (b) - Verification Methods

(a) What is a Verification Method?

A Verification Method is defined within the BCA as - "…a test, inspection, calculation or other method that determines whether a Building Solution complies with the relevant Performance Requirement."

It should be emphasised that it is not mandatory to use a prescribed BCA Verification Method. Although this is clear to regular users of the BCA, the ABCB Office still receives many 1300 calls from designers who start at the beginning of Section J and commence trying to use Verification Method JV3.

(b) What is a test?

A test verifies that a certain product or system achieves a certain performance level.

An example of a test to demonstrate compliance to a Performance Requirement would be a test to determine the actual U-value of a window.

(c) What is an inspection?

An inspection to verify whether a Building Solution satisfies a Performance Requirement could include a site inspection.

(d) What is meant by calculation?

Engineering calculations, including computer modeling, may be used to verify that a design will achieve a desired result, i.e. meet a Performance Requirement. An example of this is the calculation methodology adopted in the BCA Verification Method JV3.

(e) What is meant by another method?

This allows any other suitable method to prove that a design, construction or individual component meets a Performance Requirement. It may involve the use of fire drills to verify actual evacuation times or measuring car fume exhausts in a car park to verify a non-complying AS 1668.2 mechanical exhaust meets the Performance Requirement.

The number of possible Verification Methods can be endless depending on the situation, construction restraints and Performance Requirements to be met.

Due to the broad definition, there are many options available for use as a Verification Method. However, there must be agreement between the building designer/developer and the Building Control Authority on what Verification Method is appropriate.

Ultimately, a Verification Method provides a methodology under which a Building Solution can be assessed and generally includes a quantifiable benchmark or predetermined acceptable criteria that the solution must achieve.

There are two types of Verification Methods specified in Clause A0.9-

- the Verification Methods contained in the BCA; and
- other Verification Methods that the Building Control Authority accepts for determining compliance with the Performance Requirements.

(f) The Verification Method contained in BCA Volume One

The Verification Method contained in BCA Volume One is JV3. There was once a JV1 and JV2 but the approach in JV1 for dwellings has now been accommodated in the DTS Provisions of Part J0 while JV2, a “stated value” approach, was removed at the request of industry.

Verification Method JV3, if used, requires certain buildings to have annual energy consumption not more than had the building been built using the DTS Provisions. It is similar to the Assessment Method of “Comparison with the DTS Provisions” described later, but with certain set parameters.

(g) Other Verification Methods

Other Verification Methods, by definition, allow almost any methodology or procedure to be used to verify an Alternative Solution, particularly for a purpose built building, subject to that method being considered suitable by the Building Control Authority and used in the appropriate way.

Other Verification Methods may be adopted from overseas or may be based on JV3 but use criteria that are specific to the project rather than the more general stated criteria in JV3. In this case, the Building Control Authority would need to be certain that the building would continue to be used as proposed in the future. Such an approach would be closer to the Assessment Method “equivalent to the Deemed-to-Satisfy provisions”.

1.8.3 Assessment Method (c) - Comparison with DTS Provisions

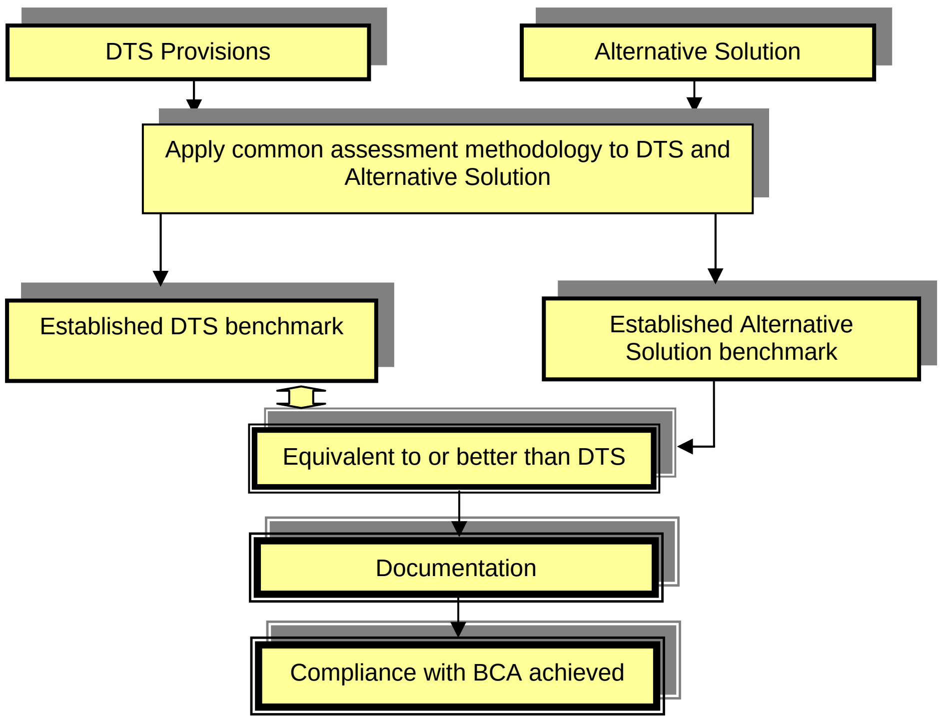
This Assessment Method involves a comparative analysis, which would demonstrate that an Alternative Solution is better or at least equivalent to the DTS Provisions. To carry out this comparison, the applicable DTS Provisions and Alternative Solutions would both need to be subjected to the same level of analysis using the same methodology. Using the DTS Provisions would provide the building designer and Building Control Authority with a defined benchmark or level for the Alternative Solution to achieve.

Following this path, it is possible to determine whether the Alternative Solution provides the same level of energy efficiency as that of the DTS Provisions. In some cases, technical analyses would be carried out using calculation methods such as a computer model.

If it is found that the Alternative Solution is equal to or better than the DTS Provision, it can be concluded that the Alternative Solution proposal satisfies the BCA Performance Requirements (see **Figure 3**). DTS Provisions means provisions which are deemed to satisfy the Performance Requirements.

Two approaches are used with energy efficiency Verification Methods. The first approach is a stated value such as a star rating to the Nationwide House Energy Rating Scheme (NatHERS) for residences or a certain MJ/m^2.annum as was used in the now discontinued JV2. The second approach is a comparative one and that is used in the current Verification Method JV3. Note that with this approach there is a degree of overlap with the Assessment Method “equivalent to the Deemed-to-Satisfy Provisions”. The specific difference is that when using JV3 stated generic input parameters must be used.

Figure 3 - Framework for establishing a benchmark to determine whether an Alternative Solution is equivalent to DTS Provisions



1.8.4 Assessment Method (d) - Expert Judgement

Expert Judgment is defined in the BCA as the judgment of an expert who has the qualifications and experience to determine whether a Building Solution complies with the BCA.

Expert Judgment is usually used when the Alternative Solution cannot be quantifiably benchmarked. It relies on a subjective opinion but by a person qualified to make a subjective assessment. In this case, the opinion should be based on certain literature, precedents or perhaps general knowledge of the issues at hand that are not taken into account in the DTS Provisions of the BCA. The person making this subjective opinion must be accepted as an expert.

In some instances, there can be a degree of overlap between Expert Judgment and other Assessment Methods. This is particularly the case with-

(a) the acceptance of documentary evidence which must comply with-

(i) BCA Clause A2.2(a)(iii) – other appropriate qualified persons; and

(ii) Clause A2.2(a)(vi) – other documentary evidence; and

(b) a comparative assessment with the DTS Provisions where only minor variation to the DTS is proposed,

The BCA defines an "expert" in the context of making Expert Judgment. The BCA definition of "Expert Judgment" is the judgment of a person who has the qualifications and experience to determine whether a Building Solution complies with the BCA Performance Requirements or DTS Provisions. Some State or Territory building law may use a different term.

1.9 Documentation

1.9.1 Evidence of Suitability

The following is an extract of Clause A2.2 for evidence of suitability.

(b) Evidence to support that a calculation method complies with an ABCB protocol may be in the form of one or a combination of the following:

(i) A certificate from a professional engineer or other appropriately qualified person which—

(A) certifies that the calculation method complies with a relevant ABCB protocol; and

(B) sets out the basis on which it is given and the extent to which relevant specifications, rules, codes of practice and other publications have been relied upon.

(ii) Any other form of documentary evidence that correctly describes how the calculation method complies with a relevant ABCB protocol.

(c) Any copy of documentary evidence submitted, must be a complete copy of the original report or document."

1.9.2 Overview of appropriate documentation

Decisions made with respect to the BCA should be fully documented. This is to ensure that the Building Control Authority is able to make an informed decision on the acceptance of the proposed Alternative Solution. The level of documentation required may vary depending on the issue. Prior to finalising a design, practitioners should discuss with the

Building Control Authority the scope of the documentation likely to be required.

1.9.3 Records to be kept

As part of a performance assessment, adequate records and documentation of the assessment should be maintained as part of the approved documentation. This is especially important for future works in that building, as any new works may affect the initial Alternative Solution that has been approved.

2. Use of Software

The use of computational software as a calculation method provides a standard and repeatable methodology and standard and repeatable values for some criteria. This standardisation is a benefit to designers, checkers and approvers as it provides familiarisation, expediency, confidence and accuracy benefits.

There is a range of building energy software used in Australia including:

- **Energy analysis software or energy simulation software**, which calculates the energy used annually by the building's energy consuming systems. It can take into account the energy used by all services and permits the operator to analyse different configurations of air-conditioning plant. This software can be used to determine a theoretical energy consumption. When determining theoretical energy consumption modelling assumptions, such as the currency of meteorological data, the occupancy hours and the internal loads, are critical. Or used to compare one building solution with another as in the BCA JV3 Verification Method. When used for comparisons, the assumptions about the currency of meteorological data, the occupancy hours and the internal loads are less critical provided the same assumptions are used in all cases being compared.
- **House Energy Rating software**, which provides a "star" rating for housing under NatHERS. It focuses on the building construction and predicts heating and cooling loads based on the energy flow through the building envelope.
- **Building Sustainability Index (BASIX)**, is a web based assessment tool that calculates the water and energy efficiency of new residential developments in NSW.
- **National Australian Built Environment Rating System (NABERS)**, managed by the NSW Department of Environment, Climate Change and Water. It measures an existing building's environmental performance during operation.
- **Green Star**, is a suite of sustainability rating tools developed by the Green Building Council of Australia. The tool considers many sustainability aspects for a range of applications and uses NABERS for the energy consumption part of the rating.

The importance of the input assumptions depends upon whether the software is for use in a comparative manner or in an absolute one. The comparative approach reduces the importance of these assumptions whereas the results under an absolute approach depend heavily on the input criteria. In the latter case the results can vary significantly depending upon the criteria used to set a target and the criteria used to meet the target. Another cause of variation is that there are different operators setting the target to those attempting to meet the target. With the comparative approach it is likely to be the same operator in both cases.

2.1 ABCB Protocols for Energy Analysis Software

For Section J, the ABCB has developed protocols for software used in energy calculations with the aim of:

- providing a legal basis for determining the suitability of particular software to demonstrate compliance with Performance Requirement JP1 via the Verification Method route;
- providing results that are repeatable and consistent to the extent reasonable; and
- being neutral to all types and sources of software in accordance with National Competition Policy.

There are two Protocols. One for House Energy rating Software that can be used for the sole-occupancy units of a Class 2 building or a Class 4 part, and another for Energy Analysis Software that can be used for a Class 3 and 5 to 9 building. The Protocol for energy analysis software describes the essential elements of software suitable for use with the energy efficiency Verification Method JV3. It also describes requirements for software development and use such as documentation, testing, quality assurance and user training.

The Protocol describes essential features of the software, specific capabilities, inputs for calculating annual energy consumption, climate data, dealing with social policy, methods of assessment, the energy analysis report, testing and quality assurance, training of users, evidence of suitability of software and process for validating and upgrading software.

2.2 Description of Energy Analysis Software

The software used for energy analysis is far more complex that that used for house energy rating and considerable time may be required to model a building and its services systems. This is because the analysis is about the energy used by the services in response to the building rather than analysing the energy flow through the building envelope. The energy used by the services will also depend upon the systems and specific equipment selected as part of the services systems. For example, a variable air volume system will have different efficiencies for a specific building to a constant volume system or a chilled beam system.

Specialised knowledge is needed to understand the various systems as well as training with the specific software. Figure 4 shows one of the input screens from a typical energy analysis software package. In this case it shows one of the configurations for an air-handling unit.

Figure 4 – Typical air-handling unit

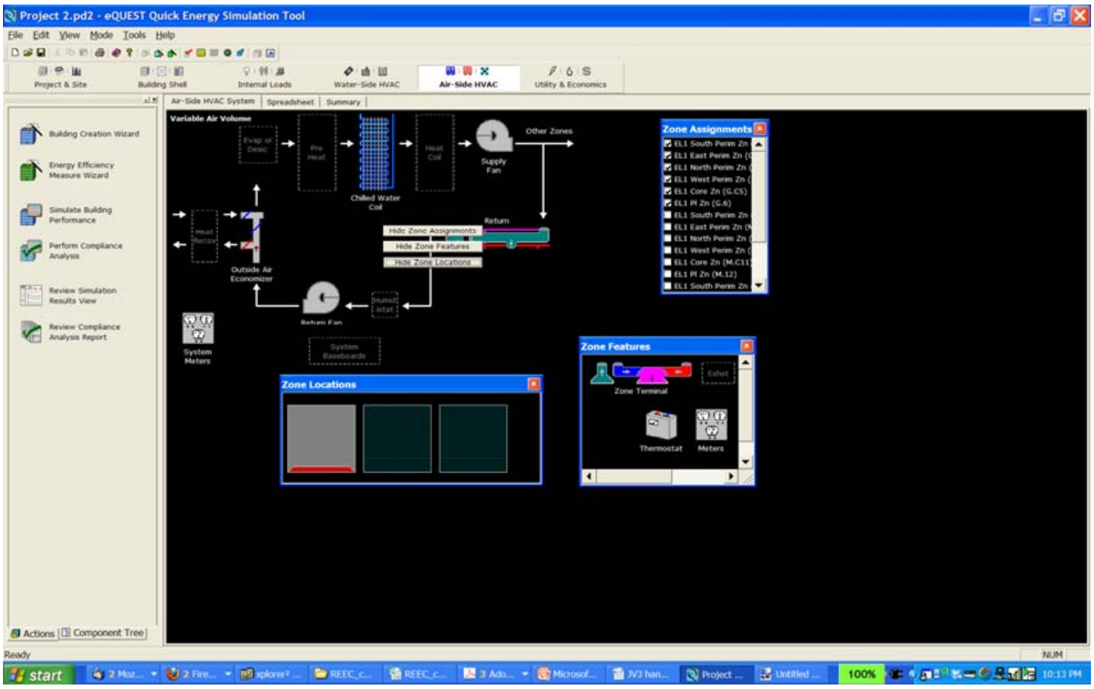
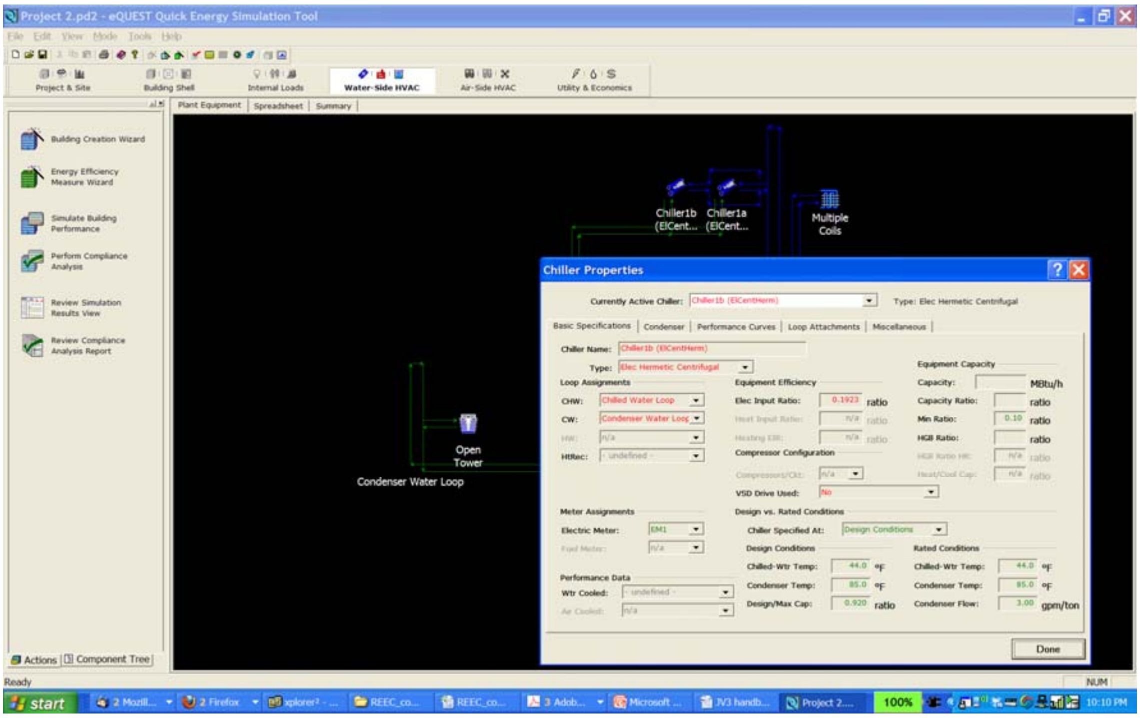


Figure 5 shows another of the input screens from a typical energy analysis software package. In this case it shows one of the configurations for a chilled water system. The operator needs to understand the differences between these systems as well as the benefits and limitations of each system.

In addition, whatever is assumed must be them included in the design such as a time clock if specific plant operation is assumed.

The software models the heat exchange between the between an air-conditioned space and the external environment to the space, hot or cold bodies in the space including people, lighting, and machines, and the air-conditioning system. The external environment includes the external ambient conditions and adjacent spaces.

Figure 5 – Typical chilled water system input screen



The heat exchange analysis includes convection to and from surfaces, radiation exchange to and from the external environment, radiation exchange between the space internal surfaces, conduction through surfaces, and changes in humidity.

These heat exchanges are interactive and are expressed within the software by mathematical relationships. To determine the amount of heating or cooling to be added to the space the solution of the heat exchanges depends on the drivers of the heat exchanges, for example, the external ambient temperature, the solar radiation, the wind velocity, the activities in the space, the previous thermal state of the building fabric and internal bodies, the reaction of the air-conditioning to the space heating or cooling demand, and the like. The power input to the air-conditioning is calculated as the air-conditioning response to the heating or cooling loads of all of the spaces.

Generally, the majority of these drivers vary with time so the power input can only be determined for the instant that the values of the drivers are set. The annual air-conditioning energy consumption is calculated as the sum of the building power calculations determined with a suitable time step between each calculation.

Historical weather data from the closest weather station, in the form of twelve months data, is used to represent the building external ambient data at the building location.

How well the modelled energy consumption represents that of the real building and its services depends on the accuracy of the modelling of the building and its services, the modelling of the interaction of the building and its services, the modelling time step chosen, how well the weather represents the weather at the location of the building and most importantly - the expertise and experience of the modeller.

2.3 Software Variations

All energy analysis software is different and although there may be similarities in the fundamental thermodynamic calculations performed, the scope and treatment of individual aspects vary from package to package. It is important that the practitioner chooses appropriate software for their project and is familiar with the limitations of the particular software being used.

2.4 Recognition of Software

The ABCB does not develop or accredit software because the BCA emphasis is on the calculation which involves both the designer and the calculation method. The choice of an appropriate calculation method and any calculations performed, are the designer's responsibility. Therefore the choice of software must be made on the basis of the appropriateness of the calculations performed by the software.

However, in order to assist the industry, the ABCB invited software suppliers to submit statements of compliance against the ABCB Protocols and those statements were provided to the State and Territory Administrations.

Administrative matters, including the possible approval of specific software for specific purposes, are the responsibility of State and Territory Administrations and Building Control Authority so advice from the respective authority should be sought.

At the time of publication, statements had been received from suppliers with respect to the following software but it should emphasised that acceptance of the statements of compliance is the State's or Territory Administration's prerogative and so it would be wise to check with your State or Territory Administration (See Appendix 1). The following is a list of software for which the supplier claimed compliance:

- BEAVER - ACADS-BSG
- DOE suite including VisualDOE, eQUEST and Energy Plus
- E20-II HAP version 4.11, 4.21
- IDA ICE v3.0
- IES Apache version 5.4.1 - Basset
- TAS - Lincolne Scott
- TRACE 700

2.5 Competence to Use the Software

To use energy rating or energy analysis software, training in the particular software is essential. The need for the software supplier to provide training is also part of the ABCB Protocols for software.

For energy rating software under the NatHERS, the user needs a basic understanding of the building elements of a house whereas for energy analysis software the user needs an understanding of the more commercial constructions and in particular a thorough understanding of the different engineering systems in a building. Most important is a thorough knowledge of the characteristics and different configurations of air-conditioning, ventilation, heating, cooling and energy reclaiming systems.

As with the software, the ABCB does not accredit users, however, some Administrations may have skill level or registration requirements while other leave recognition of this specific competence to the Building Control Authority.

2.6 Analysis Reports

The following is a typical output report from the software itself. It will need to be supplemented with the details of the Alternative Solution proposed.

Figure 6 – BEAVER printout showing the annual energy consumption

Energy Analysis Report
ACADS-BSG Program
BEAVER - Version 7.10.1

Project Name	**BEAVER Workshop**
Project Location	**Melbourne CBD**

Climatic Data

Location:	**MELBOURNE**	Date: **1971**
PROJECT NUMBER	3467	
BUILDING NAME	Bennett's Building	
ADDRESS	Office and Retail	
LOADS RUN DESC	Proposed	
SYSTEMS TITLE	VAV Chiller	
SYSTEMS RUN DESC	Proposed	
FIRM NAME	ACADS-BSG	
PROJECT ENGINEER	Bill Bloggs	
DATE	3 SEP 2009	
BUILDING CLASS	BCA Class 5 Building	

RESULTS

Calculated Annual Energy Consumption

	KWhr	MJ	%
Lighting	102943	370596	20.0
Equipment	127900	460442	24.8
Chiller Plan	76202	274327	14.8
Unitary Cooling	9910	35675	1.9
Heating Plant	68861	247898	13.4
Unitary Heating	394	1419	0.1
AHU Fan and Aux	15755	56719	3.1
Unitary Fan and Aux	6208	22349	1.2
Heating Plant Aux	2329	8383	0.5
Chiller Plant Aux	29652	106749	5.8
Ventilation	20096	72347	3.9
Domestic Hot Water	25055	90199	4.9
Lifts	8038	28939	1.6
Extra lighting	8613	31006	1.7
External Lights	13101	47162	2.5
TOTAL	**515065**	**1854233**	
per square metre	**161**	**579**	

Project floor area (m^2) 3200
Conditioned Floor Area (m^2) 2710
Condensate total (kL) 14.9

3. Verification Method JV3

The BCA 2006 energy efficiency measures introduced three new Verification Methods within the BCA but JV1 has since been relocated to Part J0 while JV2 has been withdrawn at the request of industry. The remaining Verification Method is titled JV3 (See Appendix 2).

3.1 Scope of JV3

JV3 is applicable to all Class 3, 5, 6, 7, 8 and 9 buildings. It is not applicable to Class 2 buildings and Class 4 parts of buildings. JV3 relates the energy used by the services in an Alternative Solution to the energy used by the services in a DTS solution. The BCA DTS Provisions for services in dwellings are very limited with the main focus on the dwelling's envelope.

Class 5, 6, 7, 8 and 9 buildings are more likely to be air-conditioned and artificially lit than dwellings and so it is important to consider the plant and equipment as JV3 does, rather than just the building's envelope.

3.2 Intent of JV3

The intent of any Verification Method is to demonstrate that an Alternative Solution meets the Performance Requirement.

Verification Methods can allow for innovation and better use of a building's fabric and services in order to make the building more energy efficient.

This flexibility is essential to our building environment. As a general rule, no two buildings are the same especially in terms of energy consumption. A fast food chain may have a standard design for an outlet; however, no two allotments are exactly the same and locations are climate and orientation dependant. The same building with a different orientation and exposure to the sun will, unless otherwise compensated, achieve a different level of energy consumption.

Each building is different with respect to layout, orientation, and air-conditioning. In many cases, external glazing and façade treatment is used by architects, planners and developers to provide a certain appearance that increases the usability and marketing potential of the building. However; this may be to the detriment of the energy efficiency of the building.

Verification Method JV3 allows a “trade off” between certain elements, such as a reduction in the energy efficiency of the services in the building provided there is an increase in the energy efficiency of the fabric or envelope of the building. It also permits trading between services such as between HVAC and lighting or between fabric elements such as walls and glazing. At a lower level, it also permits trading within a service such as a more efficient refrigeration chiller for a less efficient boiler, both within HVAC.

The Verification Method assesses the annual energy consumption in MJ/m^2.annum or kWh/m^2.annum of the subject building and compares it to the theoretical annual energy consumption of a reference building. The reference building annual energy consumption is that of the building as if it was constructed as a DTS compliant building.

Calculations are first carried out with the reference building in order to set a quantified benchmark, in this case the theoretical annual energy consumption. This quantified benchmark is determined using standard criteria.

The theoretical annual energy consumption of the proposed building is then calculated, using the same thermal calculation method, the same simulation methods and the same climate data as for the reference building. The theoretical annual energy consumption of the proposed building must be no greater than the DTS benchmark calculated for the reference building. This outcome must be the same with two separate runs, one with the proposed Alternative Solution and one with the alternative fabric solution defined services.

This allows flexibility in the design of the glazing (amount, quality and orientation), the fabric of the envelope (walls, floors and roof) and the services (air-conditioning, lighting, heating, etc.).

Note that under Clause A0.9 other Assessment and Verification Methods may be used as explained in Chapter 1 of this document. Their use and appropriateness is subject to approval by the Building Control Authority.

3.3 Terms Defined in the BCA

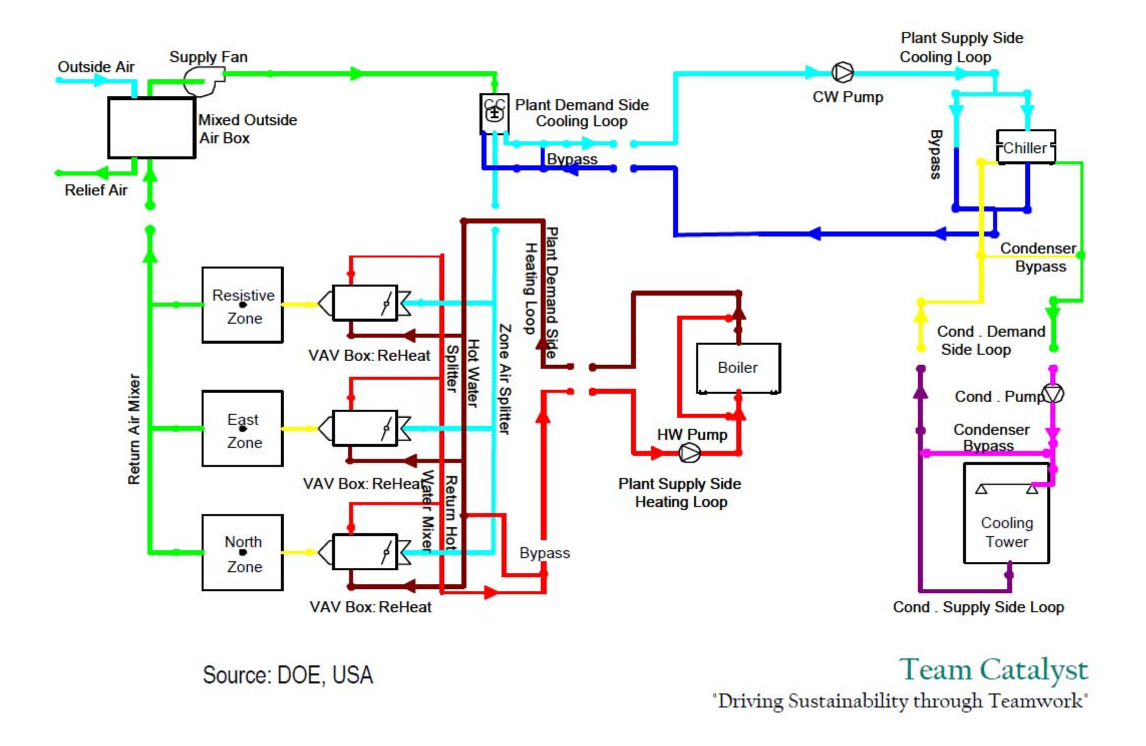
Any term in italics in the BCA is a defined term and so has a specific meaning tailored to the needs of the BCA. The following definitions are relevant in using JV3.

Air-conditioning

"**Air-conditioning** for the purposes of Section J, means a service that actively cools or heats the air within a space, but does not include a service that directly cools or heats cold rooms or hot rooms."

Figure 7 shows a typical air-conditioning system with its three distinct sub-systems, i.e. air handling, heating water and cooling water. The latter also has a further sub-system, i.e. condenser cooling water.

Figure 7 – Typical air-conditioning system



Annual energy consumption

"**Annual energy consumption** means the theoretical amount of energy used annually by the building's services, excluding kitchen exhaust and the like."

Annual energy consumption means the theoretical amount of energy used annually by the building services and includes lights and appliances even though the latter is not regulated under building law. Energy directly from gas and other fossil fuels is usually expressed as MJ/annum while electricity is expressed as kWh/annum. It is usual then to express the total as MJ/annum and then per square metre as MJ/annum.m^2.

Annual energy consumption is calculating is by using a software package that is capable of assessing the contribution of the building fabric, air infiltration, outside air ventilation, internal heat sources (people and appliances) and services such as air-conditioning systems, and artificial lighting, all specifically for the building use and location. The software must comply with the ABCB Protocol for Building Energy Analysis Software.

This theoretical amount of energy must be calculated to be consumed under certain specific assumptions including operating profiles and internal loads.

It is not considered a prediction of the actual energy consumption of an actual building as there could be major differences between the operating conditions and modelling assumptions such as the internal loads, the hours of operation and how well plant is installed, commissioned and maintained.

Annual energy consumption differs from the annual heating or cooling energy load because the consumption depends on the type of heating or cooling appliance used. For example, heating by a reverse cycle air-conditioner uses less than half the energy that a gas fired heater would use to meet the same annual energy load.

Building Classification (or "Class")

Buildings are classified as follows:

"Class 1: one or more buildings which in association constitute—

(a) **Class 1a** — a single dwelling being—

- (i) a detached house; or
- (ii) one of a group of two or more attached dwellings, each being a building, separated by a *fire-resisting* wall, including a row house, terrace house, town house or villa unit; or

(b) **Class 1b** — a boarding house, guest house, hostel or the like-

- (i) with a total area of all floors not exceeding 300 m^2 measured over the enclosing walls of the Class 1b; and
- (ii) in which not more than 12 persons would ordinarily be resident, which is not located above or below another dwelling or another Class of building other than a *private garage*.

Class 2: a building containing 2 or more *sole-occu*pancy units each being a separate dwelling.

Class 3: a residential building, other than a building of Class 1 or 2, which is a common place of long term or transient living for a number of unrelated persons, including-

(a) a boarding-house, guest house, hostel, lodging-house or backpackers accommodation; or

(b) a residential part of a hotel or motel; or

(c) a residential part of a *school*; or

(d) accommodation for the aged, children or people with disabilities; or

(e) a residential part of a *health-care building* which accommodates members of staff; or

(f) a residential part of a *detention centre*.

Class 4: a dwelling in a building that is Class 5, 6, 7, 8 or 9 if it is the only dwelling in the building.

Class 5: an office building used for professional or commercial purposes, excluding buildings of Class 6, 7, 8 or 9.

Class 6: a shop or other building for the sale of goods by retail or the supply of services direct to the public, including-

(a) an eating room, cafe, restaurant, milk or soft-drink bar; or

(b) a dining room, bar area that is not an *assembly building*, shop or kiosk part of a hotel or motel; or

(c) a hairdresser's or barber's shop, public laundry, or undertaker's establishment; or

(d) market or sale room, showroom, or *service station*.

Class 7:a building which is—

(a) **Class 7a** — a *carpark*; or

(b) **Class 7b** — for storage, or display of goods or produce for sale by wholesale.

Class 8:a laboratory, or a building in which a handicraft or process for the production, assembling, altering, repairing, packing, finishing, or cleaning of goods or produce is carried on for trade, sale, or gain.

Class 9: a building of a public nature-

(a) **Class 9a** — a *health-care building*, including those parts of the building set aside as a laboratory; or

(b) **Class 9b** — an *assembly building*, including a trade workshop, laboratory or the like in a primary or secondary *school*, but excluding any other parts of the building that are of another Class; or

(c) **Class 9c** — an *aged care building*.

Class 10:a non-habitable building or structure—

(a) **10a -** a non-habitable building being a *private garage*, carport, shed, or the like; or

(b) **Class 10b -** a structure being a fence, mast, antenna, retaining or free-standing wall, *swimming pool*, or the like."

The BCA also addresses multiple classifications under A3.3(a). If 10% or less of the floor area of a storey is used for a purpose which could be classified differently to the remainder of that storey, that part may be classified as being the same as the remainder. Laboratories and sole-occupancy units in Class 2, 3 or 4 parts are excluded from this concession.

If the provisions of A3.3(a) are used, it should be remembered that the lighting and equipment levels, people occupancy and load profiles for the area of minor use for the purposes of Section J must be in accordance with the use of the primary area.

If the storey has a very large floor area, the 10% or less concession area may also be large, even though the rest of the building is classifiable as a building which ordinarily has a lower risk potential.

The situation may arise where one section (of less than 10%) has different occupancy profiles to the main section. In this case the BCA says that for energy analysis the building may still be considered as two separate buildings. Common plant such as boilers and chillers could be apportioned. Alternatively the occupancy profiles of the main section could be used for the whole building as it will have little impact if used in both the reference building and the proposed building.

Climate zone

"**Climate Zone** means an area defined in **Figure A1.1** and in **Table A1.1** for specific locations, having energy efficiency provisions based on a range of similar climatic characteristics."

The BCA climate zones apply to the DTS Provisions and are relevant for determining the elemental provisions for the reference building. However, when modelling the Australian climate data files associated with the software are to be constructed from either Typical Meteorological Year (TMY), Weather Year for Energy Calculations (WYEC) or Test Reference Year (TRY) data files. Although reference Meteorological Year (RMY) files were not available when this Handbook was publish, this format would also be acceptable when available.

Deemed-to-Satisfy Provisions (DTS)

"**Deemed-to-Satisfy Provisions** means provisions which are deemed to satisfy the *Performance* Requirements."

These are needed for the reference building.

Envelope

"**Envelope**, for the purposes of **Section J**, means the parts of a building's *fabric* that separate a conditioned space or habitable room from—

(a) the exterior of the building; or

(b) a non-*conditioned space* including—

(i) floor of a rooftop plant room, lift-machine room or the like; and

(ii) the floor above a *carpark* or warehouse; and
(iii) the *common wall* with a *carpark*, warehouse or the like, other than a non-*conditioned space* through which conditioned air is being exhausted or relieved such as an internal corridor, cleaner's room, chemical storage room or exhaust riser."

In the BCA, this term is not limited to the building's outer shell, but also includes those continuous elements that separate a conditioned space from a non-conditioned space. For example, the floor between a plant room and an office space or the wall between a corridor and a sole-occupancy unit may be part of the envelope, rather than the outer shell. A non-conditioned space may be included within the envelope under certain circumstances.

External walls

"**External wall** means an outer wall of a building which is not a *common wall*."

Glazing

"**Glazing**, for the purposes of **Section J**, means a transparent or translucent element and its supporting frame located in the *envelope*, and includes a *window* other than a *roof light*."

The glazing definition needs to be read in conjunction with the definition of a window and roof light. It can include a glazed door. For the purposes of Section J, the glazing provides an aperture by which light and energy can flow into or from the conditioned space. Glazing includes the glass and any frame system.

Illumination Power Density

"**Illumination power density** means the total of the power that will be consumed by the lights in a space, including any lamps, ballasts, current regulators and control devices other than those that are plugged into socket outlets for intermittent use such as floor standing lamps, desk lamps or work station lamps, divided by the floor area of the space."

This term is more wide-reaching than the simpler "lamp power density" term used for residential buildings. It needs to be calculated taking account of the losses from ballast, current regulators and integral control devices associated with the lighting system including track and flexible lighting systems, and fixed lighting that is part of modular furniture and workstation lights. However, socket outlets for intermittent use such as for floor standing lamps, desk lamps, etc. are not included as it is not possible to control them through the building control process. The calculation of illumination power density does not include losses elsewhere in the system, such as in the distribution cable throughout the building.

Reference Building

"**Reference building** means a hypothetical building that is used to calculate the maximum allowable annual energy load, or maximum allowable annual energy consumption for the proposed building."

Services

"**Service**, for the purposes of Part I2 and Section J, means a mechanical or electrical system that uses energy to provide *air-conditioning*, mechanical ventilation, hot water supply, artificial lighting, vertical transport and the like within a building, but which does not include—

(a) systems used solely for emergency purposes; and
(b) cooking facilities; and
(c) portable appliances."

Services include three main items-

- air-conditioning systems that service the whole building as well as individual air-conditioning units serving the suites;
- artificial lighting for individual residential apartments as well as communal areas; and
- supply hot water systems.

These are included regardless of the energy source.

"Building" in the context of the definition of services, means the whole building for both individual units and communal facilities and public areas such as public corridors, foyers, etc.

Storey

"**Storey** means a space within a building which is situated between one floor level and the floor level next above, or if there is no floor above, the ceiling or roof above, but not—

(a) a space that contains only—
 (i) a lift *shaft*, stairway or meter room; or
 (ii) a bathroom, shower room, laundry, water closet, or other sanitary compartment; or
 (iii) intended for not more than 3 vehicles; or
 (iv) a combination of the above; or
(b) a mezzanine."

3.4 Sub-clause JV3(a) - Methodology

As described previously, the intent of JV3 is that the calculated annual energy consumption of the proposed building is not to be more than the calculated annual energy consumption were the building to be designed using the DTS Provisions. The subject building form modelled with DTS Provisions is called the “reference building”. The annual energy consumption is to be determined by a calculation method as described in the ABCB Protocol. If the energy consumption of the proposed building with the Alternative Solution does not exceed the energy consumption of the reference building, compliance with JP1 is achieved. This is required by JV3(a).

JV3 also includes provisions in JV3 (a)(ii) designed to protect the thermal performance of the building’s envelope from "trading" off its performance for over-performing services while permitting over performing envelope to be traded for underperforming services. This is achieved with an additional modelling run.

The steps to using this Verification Method are:

1. Calculate the theoretical annual energy consumption allowance by modelling a reference building, i.e. a DTS complying building based on the criteria in JV3(d)(i). This is modelling run 1.

2. Calculate the theoretical annual energy consumption of the proposed Alternative Solution (building and services) using either the subject building's criteria or those in Specification JV as required [JV3(a)(i)]. This is modelling run 2.

3. Calculate the theoretical annual energy consumption of the proposed Alternative Solution with the services modelled as if they were the same as those of the reference building [JV3 (a)(ii)]. This is modelling run 3.

4. Compare the theoretical annual energy consumption calculated in steps 2 and 3 to the annual energy consumption allowance calculated in step 1 to ensure that in both cases, the annual energy consumption of the reference building in step 1 is not exceeded by that in steps 2 and 3.

Why are two compliance runs required?

Two runs are necessary [(a)(i) and (a)(ii)], in addition to the reference building run, because if only the one run of sub-clause (a)(i) was carried out, the building may be designed to “trade-away” the thermal performance of the building’s fabric and envelope for over performing building services.

Whilst energy efficient building services are always desirable, the energy efficiency of a building’s fabric and envelope is considered to be more sustainable than that of services. Services may change over time or a lack of the necessary maintenance may cause the services to under-perform, whilst once the passive fabric energy efficiency measures are in place, they generally retain their performance for the life of the building without maintenance.

However, JV3 does permit a trade-off to go the other way. That is, the passive energy efficiency of the building's fabric and envelope may be increased in order to allow a reduction in the performance of the building services below the standard otherwise required in the DTS Provisions.

Example for the use of Verification Method JV3:

A five storey Class 5 building located in Melbourne is proposed to be assessed under Verification Method JV3. The building has insulation less than in the DTS Provisions whilst the services will have energy efficiency parameters well above the minimum of the DTS Provisions. The following calculations are made.

The annual energy consumption of the proposed building with the proposed services is calculated at 580 MJ/m^2.annum. A reference building is assessed having minimum DTS envelope characteristics as well as minimum DTS services. The annual energy consumption of the reference building and services is calculated at 620 MJ/m^2.annum.

The actual building meets the first criteria of JV3 (a) (i) for Verification Method JV3 as 580 MJ/m^2.annum is less than the 620 MJ/m^2.annum minimum.

The annual energy consumption of the proposed building is then modelled with the services at the minimum DTS standard as required by JV3 (a) (ii) and the annual energy consumption is calculated at 650 MJ/m^2.annum.

As the reference building, having minimum DTS fabric and services was previously calculated at 620 MJ/m^2.annum and the second compliance run resulted in 650 MJ/m^2.annum, the proposed building's does <u>not</u> comply under Verification Method JV3.

3.5 Sub-clause JV3 (b) – Renewable or “Free” Energy

JV3 allows renewable energy generated on-site or energy “free” from another process to be deducted from the annual energy consumption of the proposed building. This means that the “annual energy consumption” is the sum of the energy drawn annually from the electrical grid, the gas network or fuel brought in by road transport and not the total of the energy consumed by the services that use energy.

To obtain this concession, the renewable energy must be generated on-site and so Green Power, the joint initiative of ACT, NSW, SA, QLD, VIC and WA government agencies, does not get the concession.

In determining the amount of renewable energy, a designer needs to consider the likely availability of energy from the source including any down time the plant may experience for maintenance.

Free energy could include reclaimed energy from a refrigeration chiller that is used to heat water rather than being rejected through a cooling tower or energy from a process taking place in the building unrelated to the building’s services such as steam condensate from a laundry process.

3.6 Sub-clause JV3(d)(i) - Parameters for the Reference Building

A reference building is used to determine the maximum annual energy consumption allowed. This is done by applying the DTS Provisions, along with certain fixed parameters, to a proposed design. The annual energy consumption calculated then becomes the bench mark for an Alternative Solution as described in 3.4.

JV3 requires that the reference building be modelled with parameters which are considered typical for a range of buildings over their life. If these parameters were not fixed, the calculations could be manipulated by using less energy efficiency parameters when setting the allowance with the reference building.

Although the parameters may not be how the subject building is to operate, a building may change its use over its life and may even change its Classification. Different owners and different tenants will have different internal loads, different operating times and other criteria. Even though the values stated may not be those of the proposed building, they are considered reasonable averages for how some buildings operate over their life.

The reference building is that which would have been built had not the alternative innovative solution been proposed. So, for example, if there is a specific covenant such as a heritage requirement on the facade, the reference building would reflect that requirement.

These fixed standard parameters include:

(A) The DTS Provisions for Parts J1 to J7. The provisions would be the minimum or the maximum as appropriate, required by Parts J1 to J7. For example, they would include only the minimum amount of mechanical ventilation required by Part F4. It would be unreasonable to present an argument for a high outside air rate for the reference building for improved health but not the proposed building. However, it would be reasonable for the reference building to be based on the minimum requirements of Appendix A of AS 1668.2, while claiming the benefit of filtration plant allowed under that standard in the subject building. The Guide to the BCA is also useful in explaining the intent of the BCA provisions, which would apply to the reference building.

The user should also read carefully the BCA clauses and the definitions of the terms in italics. For example, when determining the maximum fan power allowance, the allowance is expressed in terms of the "*floor area* of the air-*conditioned space*" and not the whole building. Floor area in particular has a different BCA definition in different situations.

Use of the terms "system" and ""unit" in Clause J5.2(a)(vii) should also be understood. The lead-in to the clause refers to the situation where the *air-conditioning* system provides the mechanical ventilation but the specific requirement for an outside air economy cycle is about the "unit". This is because the requirement for an outside air economy cycle is typically based on the cost of the dampers and controls as against the energy saved so it relevant for a unit rather than a "system". A situation where there are 5 air-handling units each of 30 kW with their own outside air supply does not require an outdoor economy cycle but a single 150 kW system does. Likewise a system with 5 air-handling units each of 30 kW with a common outside air supply does require an outdoor economy cycle.

(B) A solar absorptance of 0.6 for external walls and 0.7 for roofs. Solar absorptance ranges from 0 (white) to 1 (black) and galvanising is around 0.55 when new but could be more when weathered.

(C) The maximum illumination power density without any increase for a control device illumination power density adjustment factor. It would be unreasonable to claim that the reference building would have had motion detectors in order to increase the illumination power density allowance. However, it would be reasonable to make allowances for Room Aspect ratios based on the room arrangement of building layout.

(D) Air-conditioning with the conditioned space temperature within the range of 18^{O}CDB to 26^{O} CDB for 98% of the plant operation time. As the same criteria must be in both the reference building and the proposed building, the exact temperature does not have a significant impact on the calculation. Where the system is a heating only one or a cooling only one, it would be reasonable to interpret this as not exceeding (above or below) the 18^{O}CDB or 26^{O} CDB for 98% which ever is applicable. Where the space is naturally ventilated it is not a conditioned space, however a space that runs in a hybrid mode will need to be modelled within the above temperature range when operating as a conditioned space.

(E) Profiles for occupancy, air-conditioning, lighting and internal heat gains from people, hot meals, appliance, equipment and hot water supply systems must be those in Specification JV unless the operating hours per year are likely to be not less than 2,500 in which case either Specification JV can be used or the likely profiles for the proposed building.

(F) Infiltration values are to be 1.0 air change per hour for a perimeter zone (of depth equal to the floor-to-ceiling height) when pressurising plant is operating, and 1.5 air change per hour for the whole building when pressurising plant is not operating. These values are consistent with those in the ASHRAE (American Society of Heating, Refrigerating & Air-conditioning Engineers) Guide.

Only the major parameters are specified because JV3 is a comparative Verification Method and so is less sensitive to input criteria provided it is the same in all runs. Alternatively, with a stated value method more input parameters would have to be specified in order to avoid a wide range of results.

3.7 Sub-clause JV3(d)(ii) - Parameters for Both Buildings

There are parameters that must be the same in both the reference building and the proposed buildings. Again, this is to avoid using energy efficiency criteria or calculations that could result in a more generous allowance using the reference building and then criteria or calculations that result in lower annual energy consumption values for the proposed building.

Those provisions that must be the same in all runs are:

(A) **The annual energy consumption calculation method itself.** Advice from industry is that different software used for energy analysis calculations can give results that may differ by up to 20% and with different operators, which would be the case if setting allowance under a stated value method, even higher. By using the same software in all runs, and in all probability the same operator doing all the runs, considerably diminishes the software differences and the operator interpretations.

(B) **The location**; being either the location where the building is to be constructed if climatic data is available, or the nearest location with similar climatic conditions in the same climate zone, for which climatic data is available. It would not be appropriate to use Wagga Wagga for the reference building and Mildura for the subject building even though they are both within the same BCA defined climate zone. This requires that the climate file that is used for the reference building will also be used for the proposed building.

(C) **Adjacent structures and features**. It would not be appropriate to treat the reference building as a Greenfield site but the proposed building as part of a campus development with other proposed buildings providing shading. Likewise in one run anticipating the demolition of a building or the growth of vegetation without doing the same in both runs.

(D) **The environmental conditions**; such as ground reflectivity, sky and ground form factors, temperature of external bounding surfaces, air velocities across external surfaces and the like. All of these aspects would be the same for both runs if the software allows them to be considered.

(E) **The building orientation**. It would not be appropriate to initially model a building with glazing facing East-West and then re orientate it so that the glazing faces North-South for the proposed solution.

(F) **The building form**; including the roof geometry, the floor plan, the number of storeys, the ground to lowest floor arrangements and the size and location of glazing. To change the form of any of these aspects could significantly change the energy consumption, particularly the glazing. The principle of the reference building is that it is the proposed building, were it designed using the DTS Provisions. For example, window sizes, number and orientation must remain the same for both runs but the SHGC and U-Value of the glazing system, and the degree of shading, can vary.

Calculating heat losses or gains through floors that are ground coupling and establishing ground temperatures under buildings is extremely difficult. Therefore heat transfer through ground coupled floors could be ignored provided it is ignored in the modelling of both the reference building and the proposed building.

(G) **The external doors.** The number and type of doors must be the same even if the software used has the ability to discriminate between the types of door and the degree of infiltration. If not, they could be omitted from all runs.

(H) **The testing standards, including for insulation, glazing, water heater and package air-conditioning equipment.** Glazing rated to Average National Average Conditions will have different performance to that rated to AFRC conditions. Likewise, insulation is rated to a different scale in the USA to AS/NZS 4859.1 ratings. Again, the principle is to use the same approach with all runs which, in the case of elements where there are specific requirements for the reference building, must be to the same requirement in the proposed building. It should be noted that the conditions under which products are rated are only for determining the rating and not the conditions in the proposed building or where the proposed building is to be located.

(I) **The thermal resistance of air films including any adjustment factors, moisture content of materials and the like.** Generally there is no reason why these values should change from one run to the next other than in an innovative solution where specific provisions are made such as revels or other such protrusions or devices used to reduce the air velocity across the external surfaces.

(J) **The dimensions of external, internal and separating walls**. If, for example, an Alternative Solution to a BCA Section C provision is proposed that would result in a reduction in wall dimensions, both the reference building and the proposed building must include the reduced dimension. Also, where tenancy fit-out layouts are not available, the same layout must be used in all runs.

(K) **The surface density of envelope walls over 220 kg/m^2**. Under JV3 it is not permitted to use, say a 220 kg/m^2 wall in the reference building and a 400 kg/m^2 wall in the Alternative Solution although it could be argued that JV3 is limited in this regard. Were a designer to want to claim the benefit of a very high mass wall it would need to be another Assessment Method or another Verification Method, possibly based on JV3.

(L) **The quality of the insulation installation.** It is not appropriate to claim "typical" or "poor" installation for the reference building and "good" installation quality for the proposed building. Were a designer to want to claim the benefit of a factory assembled panel system as against a typical site assembled wall, it would again need to be another Assessment Method or Verification Method. The Building Control Authority would need to be convinced of the improved performance, possible with test results.

(M) **Assumptions and means of calculating the temperature difference across air-conditioning zone boundaries**. Different software uses different approaches. What is important is that the same approach is used in all runs.

(N) **The floor coverings and furniture and fittings density**. Although not regulated, the amount of furniture impacts upon the energy consumption by providing a "thermal sink" which retains energy thereby reducing air-conditioning peak loads. The extent depends upon the furniture density, type, location, floor coverings and glazing coverings.

(O) **The internal shading devices, their colour and their criteria for operation**. It would not be appropriate to assume dark Venetian blinds or no blinds for the reference building but white ones closed when the sun is on that window for the proposed design. It would be appropriate to assume the blinds would be closed if the shading devices were operated manually but this is likely to have already been taken into account when the DTS glazing solution was determined.

(P) **The number and sizes of lifts and escalators and the floors served**. These are not regulated under any law (other than those for access for people with a disability) and so are determined on a commercial or waiting time basis. It would not be appropriate to vary any of these parameters between the runs but the efficiency of the lifts etc could be varied. That is, as the BCA does not have energy efficiency DTS Provisions for vertical transport, the reference building could be based on the least energy-efficient units available and the propose building could use more energy efficient ones.

(Q) **The range and type of services and energy sources other than energy generated on-site from sources that do not emit greenhouse gases such as solar and wind power**. If gas boilers are in the proposed building then oil boilers must not be used in the reference building. The exception is where in the proposed solution some or all the energy is provided by renewable energy collected or generated on-site.

(R) **The internal artificial lighting levels**. The internal artificial lighting levels should be the same in both runs and in all probability based on the recommended levels in AS/NZS 1680. Were a designer to seek to claim the benefit of lower lighting levels, say based on using some task lighting, it would need to be another Assessment Method or Verification Method and the Building Control Authority would need to be convinced of the likelihood of that philosophy continuing for the life of the building.

(S) **The internal heat gains including people, lighting, appliances, meals and other electric power loads.** As highlighted earlier, a building may change its use over its life and even its Classification. Different owners and different tenants will have different internal loads, different people density, different operating times and other criteria. Even though the values stated may not be those of the proposed building, they are considered reasonable averages for how most buildings operate over their life. Again, for a purpose built building, and subject to the Building Control Authority's agreement, another Verification Method based on JV3 could be developed using different parameters.

(T) **The air-conditioning system configuration and zones**. This is to avoid the calculations being manipulated by proposing a very basic single-zone system for the reference building and a more sophisticated variable-air-volume multi-zone system that would be considerably more efficient for the proposed building. This would effectively be a “free” allowance as most office buildings have variable-air-volume multi-zone systems. However, it is not intended to stifle innovative solutions such as chilled beams.

(U) **The daily and annual profiles of the building occupancy and operation of services**. In JV3(d)(i)(E) these parameters were set for the reference building and the same must also be used for the proposed building. These are fundamental to how the building is used and not technology based. If they were varied, it would have a major impact on the energy consumption, e.g. a 24 hour/day operating building verses a 12 hour/day operating building.

(V) **The range of internal temperatures and plant operating times**. Again, these parameters were set for the reference building in JV3(d)(i)(D) and JV3(d)(i)(E) and the same must also be used for the proposed building. However, were the plants operating times to be different in the proposed building because an energy saving feature, such as an ice storage system is proposed, then again another Verification Method based on JV3 could be developed using different operating times.

(W) **The supply hot water temperature and rate of use.** Likewise, these parameters were set for the reference building in JV3(d)(i)(E) and the same must also be used for the proposed building unless again there is innovative technology used, in which case another Verification Method could be developed.

(X) **The infiltration values**; unless there are specific additional sealing provisions or pressure testing to be undertaken with the proposed building. There is evidence of a wide range of leakage rates for buildings of different design and construction type and construction quality. As there are many construction types and construction quality is beyond the scope of the BCA, the proposed building is to have the same infiltration values as required of the reference building unless there is specific attention paid to sealing to the satisfaction of the Building Control Authority. Possible approaches could involve on-site pressure testing or leakage testing to confirm the assumptions.

(Y) **The unit capacity and sequencing for water heaters, refrigeration chillers and heat rejection equipment such as cooling towers**. As for air-conditioning plant, this is to avoid the calculations being manipulated by proposing heat rejection plant such as air cooled equipment for the reference building and then more efficient plant such as cooling towers for the proposed building. This is not to discourage any type of heat rejection equipment (which may be needed for other reasons such as Legionella control) but simply to require the same equipment to be used in all runs.

(Z) **The metabolic rate for people**. This sub-clause makes it clear that the same values are to be used in all runs. The metabolic rate is applied in JV3(d)(ii)(S) with the people density to determine the overall people load.

3.8 Sub-clause JV3(d)(iii) – Parameter for the Proposed Building

In principle, the designer can propose any solution for the proposed building and services. The only modelling constraint that applies solely to the proposed building is that the solar absorptance used for the roof and walls is to be 0.05 higher than that proposed. This is to allow for some degradation of lighter colours through weathering.

3.9 Sub-clause JV3(e) - Hot Water Supply & Vertical Transport

BCA Volume One has no energy efficiency provisions for heating supply hot water or for vertical transport (lifts or escalators). At the time of publishing this Handbook, DTS Provisions for supply water heaters had not yet been finalised although COAG had agreed upon their inclusion. Provisions for lifts and escalators have not been included because of claims by that industry that all lifts and escalators are highly efficient. The absence of these provisions means that the modeller is free to exclude these services from the modelling if they are the same in the proposed building as in the reference building.

Alternatively, as there are no DTS Provisions, the performance of a basic but realistic system can be selected for the reference building and a higher performance system selected for the proposed building. In this way a high performance hot supply water system or high performance lift can provide a credit and so go towards off-setting any under-performance of other services. While using a less energy efficient water heater for the reference building and a more energy efficient one for the proposed building may result in using a more greenhouse gas intensive energy source than is permitted under BCA 2010. However, it is intended that like space heating, the energy source for supply water heating will be restricted in BCA Volume One as it is already in BCA Volume Two.

3.10 Sub-clause JV3(f) – Attributing Lift Energy Consumption

Where a lift is included in the calculations and the lift serves more than one building classification, the energy consumption of the lift may be proportioned according to the number of storeys. This means that if four storeys of a building are retail (a Class 6 building) and 10 storey are hotel accommodation (Class 3) then 40% of the energy used by the lifts is attributed to the Class 6 building and 60% to the Class 3 building.

3.11 Sub-clause JV3(g)(i) - Implicit Requirements

JV3(g)(i) requires all aspects modelled to be achievable in the proposed building. For example, if the profile for the building says that the air-conditioning will be turned off at certain times then a time switch must be part of the solution.

3.12 Sub-clause JV3(g)(ii) – Additional Provisions

An Alternative Solution need not be restricted by any DTS Provisions. This could mean testing of components to overseas standards. However, under Verification Method JV3, sub-clause (g)(ii) certain DTS Provisions are to be complied and these include:

(A) J1.2 for general thermal construction;

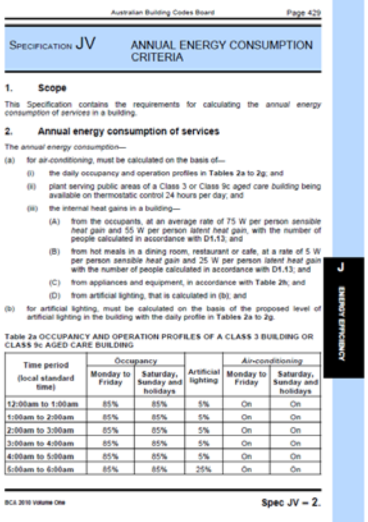
(B) J1.3(c) for compensation for a loss of ceiling insulation;

(C) J1.6(a)(ii), J1.6(c) and J1.6(d) for floor edge insulation;

(D) BS 7190 for testing a water heater;

(E) AS/NZS 3823.1.2 for testing package *air-conditioning* equipment; and

(F) ARI 550/590 for testing a refrigeration chiller (reference under review).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Specification JV ANNUAL ENERGY CONSUMPTION CRITERIA

1. **Scope**

This Specification contains the requirements for calculating the *annual energy consumption of services* in a building.

2. **Annual energy consumption of services**

The *annual energy consumption*—

(a) for *air-conditioning*, must be calculated on the basis of—

(i) the daily occupancy and operation profiles in **Tables 2a to 2g**; and

(ii) plant serving public areas of a Class 3 or Class 9c *aged care building* being available on thermostatic control 24 hours per day; and

(iii) the internal heat gains in a building—

(A) from the occupants, at an average rate of 75 W per person *sensible heat gain* and 55 W per person *latent heat gain*, with the number of people calculated in accordance with D1.13; and

(B) from hot meals in a dining room, restaurant or cafe, at a rate of 5 W per person *sensible heat gain* and 25 W per person *latent heat gain* with the number of people calculated in accordance with D1.13; and

(C) from appliances and equipment, in accordance with **Table 2h**; and

(D) from artificial lighting, that is calculated in (b); and

(b) for artificial lighting, must be calculated on the basis of the proposed level of artificial lighting in the building with the daily profile in **Tables 2a to 2g**.

Table 2a OCCUPANCY AND OPERATION PROFILES OF A CLASS 3 BUILDING OR CLASS 9c AGED CARE BUILDING

Time period (local standard time)	Occupancy		Artificial lighting	*Air-conditioning*	
	Monday to Friday	Saturday, Sunday and holidays		Monday to Friday	Saturday, Sunday and holidays
12:00am to 1:00am	85%	85%	5%	On	On
1:00am to 2:00am	85%	85%	5%	On	On
2:00am to 3:00am	85%	85%	5%	On	On
3:00am to 4:00am	85%	85%	5%	On	On
4:00am to 5:00am	85%	85%	5%	On	On
5:00am to 6:00am	85%	85%	25%	On	On

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3.13 Specification JV – Annual Energy Consumption Criteria

Specification JV provides details of criteria for use in calculating the annual energy consumption of services. Although the values stated may not be those actually achieved in some buildings, they are considered the most typical over a range of occupancies. They only have to be used as part of the Verification Method if the annual operating hours of the proposed building are less than 2,500 hours [see JV3(d)(i)(E)]. If the operating hours per year are 2,500 or more, the modeller can either use the criteria in Specification JV or the expected profiles of the proposed building. The greater the number of operating hours, the less is the impact of different values provided the same are used in all runs.

Specification JV contains:

- Daily occupancy and operating profiles for a Class 3 building, a Class 5 building, a Class 6 shop, shopping centre, restaurant or cafe, a Class 8 laboratory, a Class 9a clinic, day surgery or procedure unit, a Class 9a ward area, a Class 9b theatre, cinema or school and a Class 9c aged care building. These include-
  - occupancy starting and finishing times
  - artificial lighting percentage operating
  - air-conditioning operating
- Internal heat gains from appliances and equipment
- Hot water supply consumption rates

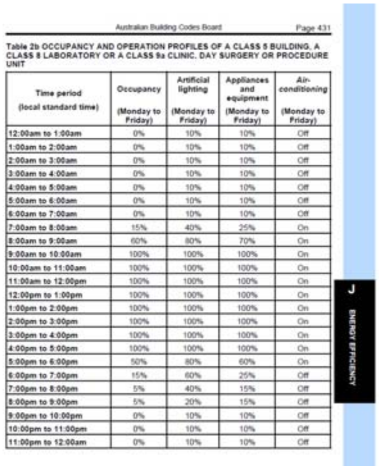

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Table 2b OCCUPANCY AND OPERATION PROFILES OF A CLASS 5 BUILDING, A CLASS 8 LABORATORY OR A CLASS 9a CLINIC, DAY SURGERY OR PROCEDURE UNIT

Time period (local standard time)	Occupancy (Monday to Friday)	Artificial lighting (Monday to Friday)	Appliances and equipment (Monday to Friday)	Air-conditioning (Monday to Friday)
12:00am to 1:00am	0%	10%	10%	Off
1:00am to 2:00am	0%	10%	10%	Off
2:00am to 3:00am	0%	10%	10%	Off
3:00am to 4:00am	0%	10%	10%	Off
4:00am to 5:00am	0%	10%	10%	Off
5:00am to 6:00am	0%	10%	10%	Off
6:00am to 7:00am	0%	10%	10%	Off
7:00am to 8:00am	15%	40%	25%	On
8:00am to 9:00am	60%	80%	70%	On
9:00am to 10:00am	100%	100%	100%	On
10:00am to 11:00am	100%	100%	100%	On
11:00am to 12:00pm	100%	100%	100%	On
12:00pm to 1:00pm	100%	100%	100%	On
1:00pm to 2:00pm	100%	100%	100%	On
2:00pm to 3:00pm	100%	100%	100%	On
3:00pm to 4:00pm	100%	100%	100%	On
4:00pm to 5:00pm	100%	100%	100%	On
5:00pm to 6:00pm	50%	80%	60%	On
6:00pm to 7:00pm	15%	60%	25%	Off
7:00pm to 8:00pm	5%	40%	15%	Off
8:00pm to 9:00pm	5%	20%	15%	Off
9:00pm to 10:00pm	0%	10%	10%	Off
10:00pm to 11:00pm	0%	10%	10%	Off
11:00pm to 12:00am	0%	10%	10%	Off

J ENERGY EFFICIENCY

It is not practical to have occupancy and equipment operation profiles for all possible uses of buildings. Tables 2a to 2g are for the most common applications. Those for office buildings are the same as those used in the Australian Building Greenhouse Rating Scheme (ABGR) with respect to people and air-conditioning but not for lighting and office equipment. The ABGR has higher values for equipment use. Others values are based on those in the ASHRAE (American Society of Heating, Refrigerating & Air-conditioning Engineers) Guide.

APPENDIX 1 – List of State & Territory Administrations

The following are the contact details for the ABCB and each State and Territory Administration as of 1 May 2010:

AUSTRALIAN BUILDING CODES BOARD	**Telephone:** 1300 134 631 **E-mail:** BCA@abcb.gov.au **Web site:** www.abcb.gov.au
AUSTRALIAN CAPITAL TERRITORY ACT Planning and Land Authority GPO Box 1908 Canberra ACT 2601	**Telephone:** 02 6207 1923 **E-mail:** actpla.customer.services@act.gov.au **Hours:** 8.30am-4.30pm **Web site:** www.actpla.act.gov.au
NEW SOUTH WALES Department of Planning Lands Department Building GPO Box 39, Sydney NSW 2001	**Telephone:** 02 9228 6111 **Hours:** 9.30am-11.30am (Tuesday, Wednesday, Thursday) **Web site:** www.planning.nsw.gov.au
NORTHERN TERRITORY Department of Lands and Planning GPO Box 1680, Darwin NT 0801	**Telephone:** 08 8999 8985 **E-mail**: bas.lpe@nt.gov.au **Hours:** 8.00am-4.00pm **Web site:** www.nt.gov.au
QUEENSLAND Building Codes Queensland Department of Infrastructure and Planning PO Box 15009, City East QLD 4002	**Telephone:** 07 3239 6369 **E-mail:** buildingcodes@dip.qld.gov.au **Hours:** 8.00am-5.00pm **Web site:** www.dip.qld.gov.au

SOUTH AUSTRALIA Department of Planning and Local Government Building Policy Branch GPO Box 1815 Adelaide SA 5001	**Telephone:** 08 8303 0602 **E-mail:** plnsa.building@saugov.sa.gov.au **Hours:** 8.30am-5.00pm **Web site:** www.planning.sa.gov.au
TASMANIA Department of Justice Workplace Standards Tasmania Building Control Branch PO Box 56 Rosny Park TAS 7018	**Telephone:** 03 62337657 **E-mail:** wstinfo@justice.tas.gov.au **Hours:** 9.00am-5.00pm **Web site:** www.wst.tas.gov.au
VICTORIA Building Commission Victoria PO Box 536, Melbourne VIC 3001	**Telephone:** 1300 815 127 **E-mail:** technicalenquiry@buildingcommission.com.au **Hours:** 8.30am-5.00pm **Web site:** www.buildingcommission.com.au
WESTERN AUSTRALIA Building Commission, Department of Commerce Locked Bag 12 West Perth W.A. 6872	**Telephone:** 08 9476 1333 **E-mail:** info@buildingcommission.wa.gov.au **Hours:** 8.30am-5.00pm **Web site:** www.buildingcommission.wa.gov.au

APPENDIX 2 – JV3 Text

JV3 Verification using a reference building

(a) For a Class 3, 5, 6, 7, 8 and 9 building, compliance with JP1 is verified when it is determined that the *annual energy consumption* of the proposed building with its *services* is not more than the *annual energy consumption* of a *reference building* when—

(i) the proposed building is modelled with the proposed *services*; and

(ii) the proposed building is modelled with the same *services* as the *reference building*.

(b) The *annual energy consumption* of the proposed building in (a) may be reduced by the amount of energy obtained from—

(i) a source that is renewable on-site such as solar, geothermal or wind; or

(ii) another process as reclaimed energy.

(c) The *annual energy consumption* calculation method must comply with the ABCB Protocol for Building Energy Analysis Software.

(d) The *annual energy consumption* in (a) must be calculated—

(i) for the reference building, using—

(A) the DTS Provisions for Parts J1 to J7 but including only the minimum amount of mechanical ventilation required by Part F4; and

(B) a solar absorptance of 0.6 for external walls and 0.7 for roofs; and

(C) the maximum illumination power density without any increase for a control device illumination power density adjustment factor; and

(D) air-conditioning with the conditioned space temperature within the range of 18° CDB to 26° CDB for 98% of the plant operation time; and

(E) the profiles for occupancy, air-conditioning, lighting and internal heat gains from people, hot meals, appliance, equipment and hot water supply systems—

(aa) of the actual building provided the operating hours per year are not less than 2,500; or

(bb) of Specification JV; and

(F) infiltration values—

(aa) for a perimeter zone of depth equal to the floor-to-ceiling height, when pressurising plant is operating, 1.0 air change per hour; and

(bb) for the whole building, when pressurising plant is not operating, 1.5 air change per hour; and

(ii) for both the proposed building and the *reference building* using the same—

(A) *annual energy consumption* calculation method; and

(B) location, being either the location where the building is to be constructed if appropriate climatic data is available, or the nearest location with similar climatic conditions in the same *climate zone*, for which climatic data is available; and

(C) adjacent structures and features; and

(D) environmental conditions such as ground reflectivity, sky and ground form factors, temperature of external bounding surfaces, air velocities across external surfaces and the like; and

(E) orientation; and

(F) building form, including—

(aa) the roof geometry; and

(bb) the floor plan; and

(cc) the number of storeys; and

(dd) the ground to lowest floor arrangements; and

(ee) the size and location of *glazing*; and

(G) external doors; and

(H) testing standards including for insulation, *glazing*, water heater and package *air-conditioning* equipment; and

(I) thermal resistance of air films including any adjustment factors, moisture content of materials and the like; and

(J) dimensions of external, internal and separating walls; and

(K) surface density of *envelope* walls over 220 kg/m^2; and

(L) quality of insulation installation; and

(M) assumptions and means of calculating the temperature difference across *air-conditioning* zone boundaries; and

(N) floor coverings and furniture and fittings density; and

(O) internal shading devices, their colour and their criteria for operation; and

(P) number, sizes and floors served by lifts and escalators; and

(Q) range and type of *services* and energy sources other than energy generated on-site from sources that do not emit greenhouse gases such as solar and wind power; and

(R) internal artificial lighting levels; and

(S) internal heat gains including people, lighting, appliances, meals and other electric power loads; and

(T) *air-conditioning* system configuration and zones; and

(U) daily and annual profiles of the—

(aa) building occupancy; and

(bb) operation of *services*; and

(V) range of internal temperatures and plant operating times; and

(W) supply hot water temperature and rate of use; and

(X) infiltration values unless there are specific additional sealing provisions or pressure testing to be undertaken; and

(Y) unit capacity and sequencing for water heaters, refrigeration chillers and heat rejection equipment such as cooling towers; and

(Z) metabolic rate for people; and

(iii) for the proposed building using a solar absorptance for the roof and walls 0.05 higher than that proposed; and

(e) Where the *annual energy consumption* of the hot water supply or the lifts and escalators are the same in the proposed building and the *reference building*, they may be omitted from the calculation of both the proposed building and the *reference building*.

(f) A lift in a building with more than one classification may be proportioned according to the number of *storeys* of the part for which the *annual energy consumption* is being calculated.

(g) The design must include—

(i) the ability to achieve all the criteria used in the *annual energy consumption* calculation method such as having an automatic operation controlling device capable of turning lighting, and *air-conditioning* plant on and off in accordance with the occupancy and operating profiles used; and

(ii) compliance with—

(A) J1.2 for general thermal construction; and

(B) J1.3(c) for compensation for a loss of ceiling insulation; and

(C) J1.6(a)(ii), J1.6(c) and J1.6(d) for floor edge insulation; and

(D) BS 7190 for testing a water heater; and

(E) AS/NZS 3823.1.2 at test condition T1 for testing package *air-conditioning* equipment; and

(F) ARI 550/590 for testing a refrigeration chiller.

APPENDIX 3 – ABCB Protocol for Building Energy Analysis Software Version 2006

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Foreword

The Australian Building Codes Board (ABCB), in conjunction with State and Territory building control Administrations, is responsible for developing and maintaining the provisions of the Building Code of Australia (BCA), including those for energy efficiency.

The BCA is given legal status by the State and Territory Building Acts and Regulations. Any material referenced in the BCA needs to be clearly identified and described as it also has legal status under those Acts and Regulations.

The main text of this Protocol has been prepared for referencing by the BCA. Other matters that merit attention, but which are not suitable for referencing, are contained in this Foreword.

Need for a Protocol

Energy analysis software has been used for many years by designers, energy auditors and energy assessors to assess the energy consumption of a building. Software packages vary considerably in scope, complexity and application and therefore can deliver a wide range of results.

To ensure that software used to demonstrate compliance with the BCA energy efficiency measures is of an appropriate standard; this Protocol defines minimum requirements for software and training of its users. It also provides a process for demonstrating the acceptability of new software and revisions to existing software.

BCA Energy Efficiency Measures

The BCA energy efficiency measures include three Verification Methods that can be used to demonstrate that an Alternative Solution complies with the BCA Performance Requirement. It is anticipated that software will be used to assist in demonstrating compliance by evaluating the annual energy consumption of a particular design.

The first Verification Method is for assessing the energy performance of housing and other private dwellings (i.e. Class 1 and 10 buildings, the sole-occupancy units of Class 2 buildings and Class 4 parts). It is based on achieving a star rating to the Nationwide House Energy Rating Scheme. The minimum requirements for energy rating software used to assess these buildings are described in a separate BCA reference document titled "Protocol for House Energy Rating Software".

The two other Verification Methods are based on the alternative building solution consuming the same or less energy than a maximum amount. In one method, the maximum amount is stated, and in the other, it is calculated. In both cases the amount equates to what a building would use under the defined conditions with the Deemed-to-Satisfy Provisions. Either method can be used for buildings (other than Class 1 and 10 buildings, the sole-occupancy units of Class 2 buildings and Class 4 parts), although the stated amount method is restricted to the most common buildings uses.

Aims of this Protocol

The aims of this Protocol are to:

- provide a legal basis for determining the suitability of particular software to demonstrate compliance with BCA Performance Requirement JP1 via the Verification Method route;
- provide results that are repeatable and consistent using different software; and
- be neutral to all types and sources of software in accordance with National Competition Policy.

Suitability of software

Software suppliers or their agents may need to provide an assurance, with supporting evidence, that their software complies with this Protocol. Accepting this assurance is the responsibility of the appropriate authority. Any assurances the ABCB Office may receive are forwarded to the State and Territory building control Administrations.

Process for revising the Protocol

The Protocol may be revised from time to time as necessary. Revisions will occur in consultation with the State and Territory building control Administrations.

Other matters

Additional requirements that are not included in the Protocol but are necessary to assure the reliability of modelling outcomes are:

Testing that incorporates a procedure for rectifying software faults and inaccuracies.

Instructions for the use of the software, including:

- general software operating instructions and procedures (how to input the data) for all required building scenarios; and
- details of all software functions, settings and limitations.

Software support including:

- support for software users, explanation of procedures, documentation of all technical limitations, and a help service to provide technical and functional information to users and other interested parties; and
- advice to users on the appropriateness of manipulating the software beyond its stated use; and
- a procedure for publishing and disseminating updates not in the original software documentation (including both new capabilities and new library data files).

Version control of the software as part of the quality assurance program.

State & Territory regulatory matters

Some States or Territories may have additional requirements for software in order to provide assurance and demonstrate reliability of modelling outcomes. These may include:

- **Accreditation of users**, possibly including the passing of an examination and registration.

- **Contact details** for the Software Company, or agent, responsible for the software including matters relating to software validation, support, testing, documentation and updates.
- **Software validation** and a quality assurance framework for integrating feedback and concerns of software accuracy.
- **Documentation for any State/Territory based:**
  - examination schemes that States, Territories or industry may require for the examination of candidates in the use of software; and
  - auditing schemes that States or Territories may require for ensuring that a sufficient sample of modelling runs be carried out as an audit for quality control of accredited assessors.

Energy analysis report format

An energy analysis report is to be prepared by the energy analyst to quantify the modelled energy consumption under certain conditions and in so doing, assist in demonstrating compliance of an alternative building solution. Appendix A has been developed to provide an indication of the information needed in an energy analysis report. This sample is intended to provide sufficient information to facilitate the work of the analyst's supervisor by identifying key inputs and outputs for confirmation.

It could also be extended or supplemented to provide the regulatory information needed by the building control authority by identifying items that can be physically verified, the elements being varied in the alternative solution and the particular BCA assessment method or Verification Method used.

The information described in the Appendix will also help in providing consistency of the inputs and outputs of the main energy analysis program and other supporting programs.

PROTOCOL FOR BUILDING ENERGY ANALYSIS SOFTWARE

1. Scope

This Protocol describes the essential elements of software suitable for use primarily with the energy efficiency Verification Methods JV2 and JV3 of the Building Code of Australia Volume One. It also describes requirements for software development and use such as documentation, testing, quality assurance and user training.

2. Purpose and context of use

This Protocol has been developed to specify the requirements of energy software that is used for calculating the annual energy consumption of a building in accordance with the BCA energy efficiency measures. Software in accordance with this Protocol can be used to demonstrate compliance with Performance Requirement JP1 via one of the two Verification Methods JV2 or JV3.

3. Essential features of the software

To comply with this Protocol, software must:

- be commercially available; and
- be based on a simulation program with an hourly climate data file; and
- be capable of computing the annual energy consumption of a building in accordance with the Verification Methods of the BCA; and
- be capable of geometrically describing the building in three dimensions including taking account of surface azimuth, tilt angle and adjacent structures and features; and
- provide results comparable with other similar software in accordance with ASHRAE Standard 140 -2001 Standard Method of Test for the Evaluation of Building Energy Analysis Computer programs using the International Energy Agency BESTEST (see Clause 8); and
- address all the specific capabilities in clause 4.

4. Specific capabilities

The software must be capable of addressing all the specific aspects of BCA Verification Methods JV2 and JV3, Specification JV and Clauses 4.1 and 4.2 below, by either direct modelling or by adding in pre-determined data.

The software must use the values for the thermal properties of building products that are available in Australia, either by accessing an extensive library of local products or by appropriately modifying in-built values.

4.1 By direct modelling

Aspects of thermal modelling that the software must be capable of addressing directly are:

- the energy flow through the building's envelope, including at adiabatic surfaces and also including thermal storage effects;
- accurately modelling the performance of the air-conditioning and ventilation systems, including plant and equipment using their energy input ratios, coefficients of performance, or efficiency at full and part load;
- the control strategies, sequencing of plant and equipment, controlled settings and types of controls;
- the design relative humidity range; and
- the different energy types, e.g. electricity, gas, oil.

4.2 By direct modelling or by adding in pre-determined data

Aspects of thermal modelling that may be addressed by adding in pre-determined data rather than by direct modelling are:

- lighting systems and equipment, provided the calculation included consideration of their loads, operating profiles and the distribution of the lighting load between the space load and return air load;
- vertical transport loads; and
- supply hot water loads in accordance with BCA Specification JV.

5. Inputs for calculating annual energy consumption

5.1 Climate Data

Climate data used must be based on hourly data derived from Australian meteorological records taken at no more than 3 hourly intervals and adjusted to provide a representative year for the proposed locations (such as Test Reference Year, Typical Meteorological Year or Weather Year for Energy Calculations).

Where sufficient records are unavailable, the data needed may be estimated from other recorded data provided a reliable method is used to make these estimates, e.g. cloud cover records or satellite measurements can be used to estimate solar radiation data in the absence of recorded solar data.

Appropriate climate data based on the Australian Bureau of Meteorology records is available in the "Australian Climatic Data Bank for Use in the Estimation of Building Energy Use" which is maintained by ACADS-BSG by agreement with the Australian Government.

5.2 Other technical inputs

Program inputs for calculating the annual energy consumption of services in a building must be in accordance with Specification JV in BCA Volume One.

Program inputs for calculating the annual energy consumption of a reference building must be in accordance with Verification Method JV3.

5.3 Social policy

The software and its recorded outputs must not contain computations of a social policy nature. Computations of a social policy nature include inputs or settings based on a policy decision rather than engineering principles (eg, adding a weighting to the annual energy consumption on the basis of the floor area of the building).

6. Methods of assessment

The BCA energy efficiency Verification Methods JV2 and JV3 are available as a means for assessing compliance with Performance Requirement JP1. The definition of the nominated thermal calculation method permits the use of software.

Any software used in the Verification Methods must be based on well-established models that are in accordance with the principles of thermodynamics and fluid mechanics. The calculation methodology used in the software must be documented and be available for inspection.

Sources of reference data on the thermal properties of building materials, insulation etc. must be identified and be from test results or authoritative data sources such as the Australian Institute of Refrigeration, Air-Conditioning and Heating (AIRAH), the American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) or other recognised international standard.

7. Energy analysis report

The energy analysis report must include all relevant inputs for the building fabric, the air-conditioning and ventilation systems, the lighting and power systems, the vertical transport systems and the supply hot water systems.

Inputs and outputs must be detailed on the energy analysis report produced in order to demonstrate compliance with the chosen Verification Method and so that in conjunction with the relevant plans and specifications and any supplementary regulatory information, the building control authority can check compliance with JV2 or JV3 including Specification JV. Appendix A provides a sample energy analysis report format.

The outputs must be presented in terms of annual energy consumption of the building in MJ/m2 of floor area per annum for a particular climate region for both (a)(viii) and (a)(ix) of

JV2 and JV3. A distinction must be made as to whether the energy source for the building heating is electricity (with heat pump plant) or gas.

The energy analysis report must also include details of any limitations of the software or any approximations that were made to adapt the software to the application.

8. Testing and quality assurance

The software must be tested in accordance with ASHRAE Standard 140 -2001 'Standard Method of Test for the Evaluation of Building Energy Analysis Computer Programs' using the International Energy Agency BESTEST. The results should be within the range of results from acceptable comparable programs indicated in the Standard.

While results that fall outside this range are not necessarily incorrect, the sources of the differences must be investigated, documented and made known, particularly to the building control authority.

The software supplier must have in place a quality assurance program and be able to demonstrate its effectiveness.

9. Training of users

A training program for users must be available. This program must include training in the current version and any proposed new version of the software. Trainers must be technically qualified and be well versed in the functionality of the particular program and the calculation methods employed.

10. Evidence of suitability of software

Evidence must be produced to demonstrate that the software is suitable. This will include evidence that:

a) the software has the features outlined in Clause 3 and the specific capabilities outlined in Clause 4; and

b) the software has undergone appropriate testing and result analysis, and the process has undergone quality assurance; and

c) a training program is available for users.

Evidence of training must state the software name and the version.

The status of the software, such as whether it has been approved by any appropriate authority, must also be clearly indicated.

11. Process for validating and upgrading software

Energy analysis software used to demonstrate compliance with the BCA Volume One Performance Requirement JP1 must meet the requirements of this Protocol.

The software providers are responsible for validating software and correcting deficiencies and faults.

The ABCB should be advised, in writing, of any new validated versions of the software and the corrections or revisions to the software to ensure that building control authorities, and in turn the practitioners are adequately informed. Correspondence should be sent to:

The General Manager

Australian Building Codes Board

GPO Box 9839

CANBERRA ACT 2601

Any revisions, updates or new versions must be identified by a unique number or other form of designation. The status of any revisions, updates or new versions, such as whether it has been approved by an appropriate authority, must also be clearly indicated.

APPENDIX A - SAMPLE ENERGY ANALYSIS REPORT

GENERAL

Reference No:...Date:...............................

Property title:...

Address:..

Building Class and use:...

Verification Method used (JV2 or JV3):..

Location used (if JV2 see Table JV2 of BCA Volume 1):..

BCA Climate Zone (if for JV3 reference building):...

Name and version of the software used in analysis:..

Name and contact details of the organisation responsible for the analysis:

....

....

Name, qualifications and training with the software of the person responsible for the analysis:

....

OUTPUT

Temperature control

The percentage of the plant operating time that the temperature can be maintained within the required range [Spec. JV Clause 2(a)(i)]...(%)

Annual energy consumption

Floor area used in calculating annual energy consumption:………..…………..……(m^2)

Energy source for heating:..

Components of annual energy consumption

Component	Calculated annual energy consumption for each run			
	With proposed building fabric and services (a)(i)		With proposed building fabric and specified services (a)(ii)	
	Electricity	**Gas**	**Electricity**	**Gas**
Lighting	kWh	MJ	kWh	MJ
Heating	kWh	MJ	kWh	MJ
Cooling	kWh	MJ	kWh	MJ
Air-handling	kWh	MJ	kWh	MJ
Ventilation	kWh	MJ	kWh	MJ
Lifts	kWh	MJ	kWh	MJ
Hot water supply	kWh	MJ	kWh	MJ
Sub-total	kWh	MJ	kWh	MJ
Conversion factor	times 3.6		times 3.6	
Conversion	MJ		MJ	
Total	**MJ**		**MJ**	

Annual energy consumption allowance.. (MJ/m^2.annum)

Annual energy consumption calculated (a)(viii)...................................... (MJ/m^2.annum)

Annual energy consumption calculated (a)(ix).. (MJ/m^2.annum)

INPUTS

(a) For air-conditioning

Ground floor construction:..

External surfaces solar absorptance:..

R-Value of internal air film:..

External shading:..

Glazing area distribution:...

Solar radiation value at which blinds operate:...

Space temperature range:..

Furniture and fittings density:...

Daily occupancy and operating profile:..

Sensible internal heat gain per occupant:...

Latent internal heat gain per occupant:...

Occupant density:...

Air-conditioning system selection:..

Availability of plant:..

Reheat limit:...

Outside air cycle:...

Mechanical ventilation rate - outside air:..

Mechanical ventilation rate - exhaust air:...

Exhaust ventilation system operation:...

Internal heat gains from appliances and equipment:...

Internal heat gains from artificial lighting:...

Infiltration air change rate per hour when pressurisation plant operating:.........................

Infiltration air change rate per hour when pressurisation plant not operating:...................

How heat migration across air-conditioning zone boundaries has been assessed:

....

Carpark contaminant control:..

....

Pump speed control:...

Plant sequencing and flow water control:..

Boiler or heating water heater efficiency:...

Package air-conditioning plant energy efficiency ratio:...

Refrigerant chiller part load energy efficiency ratio:...

Air cooled condenser fan performance and control:..

Cooling tower fan performance and control:..

(b) For lighting

Average lighting levels:...

Light power density:...

Lighting controls:...

(c) For lifts and escalators

....

(d) For hot water supply

....

(e) Any concession for energy generated on site such as from solar or wind power

....

(f) Description of procedures used for those inputs not directly available in the software and needing manipulation

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Note: This energy analysis report should be read in conjunction with the relevant plans and specifications and any supplementary regulatory information.